

RKNN3 Supported Operators and Constraints Reference

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Preface

RKNN3 Supported Operators and Constraints Reference

Intended Audience

This document (this guide) is mainly intended for:

Technical support engineers

Software development engineers

Revision History

Version	Author	Date	Change Description
v1.1.0	NPU Team	2026-08-14	1. Initial release of RK18X/RK3572 platform operator support and constraint reference

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RK182X

Convolution

Conv

Convolution computation.

FORMULA

```
output = Conv(input_tensor, weight, bias)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::_convolution
```

INPUTS

- **input_tensor** - `T1` (required):
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
- **weight** - `T1` (required):
 - Shape: OIHW[output_channel, input_channel, kernel_h, kernel_w]
 - Quantization: per-layer, per-channel
 - kernel_h range: [1, 31]
 - kernel_w range: [1, 31]
- **bias** - `T1` (optional):
 - Shape: C[output_channel]
 - Quantization: per-layer

OUTPUTS

- **output** - `T1`:
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

ATTRIBUTES

- **strides** - `int[stride_h, stride_w]`:

- stride_h range: [1, 7]
- stride_w range: [1, 7]
- **pads** - `int[pads_top, pads_left, pads_bottom, pads_right]`:
 - pads_top range: [0, 15]
 - pads_left range: [0, 15]
 - pads_bottom range: [0, 15]
 - pads_right range: [0, 15]
- **group** - `int`: —
- **dilations** - `int[dilation_h, dilation_w]`:
 - dilation_h range: [1, 31]
 - dilation_w range: [1, 31]

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<float16>`):

SPECIAL CASES

Depthwise Convolution When Conv is used with depthwise convolution group semantics, the following differential specifications apply.

- **Input Differences** — **weight**: kernel_h range: [1, 8]; kernel_w range: [1, 8]
- **Attribute Differences** — **dilations**: dilation_h range: [1, 32]; dilation_w range: [1, 32]

ConvTranspose

Transposed convolution computation.

FORMULA

```
output = ConvTranspose(input_tensor, weight, bias)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::conv_transpose1d, aten::conv_transpose2d
```

INPUTS

- **input_tensor** - `T1` (required):
 - Shape: NCHW[batch, channel, height, width]

- Quantization: per-layer
- When `dilated_kernel_h > 1`, `input_tensor.width < 16383`.
The first layer input width must also satisfy the model input limitations.
- **weight** - `T1` (required):
 - Shape: OIHW[output_channel, input_channel, kernel_h, kernel_w]
 - Quantization: per-layer, per-channel
 - kernel_h range: [1, 31]
 - kernel_w range: [1, 31]
- **bias** - `T1` (optional):
 - Shape: C[output_channel]
 - Quantization: per-layer

OUTPUTS

- **output** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

ATTRIBUTES

- **strides** - `int[stride_h, stride_w]`:
 - stride_h range: [1, 8]
 - stride_w range: [1, 8]
- **pads** - `int[pads_top, pads_left, pads_bottom, pads_right]`:
 - pads_top range: [0, 15]
 - pads_left range: [0, 15]
 - pads_bottom range: [0, 15]
 - pads_right range: [0, 15]
 - `kernel_h * dilation_h - dilation_h - pads_top < 0` is not supported
 - `kernel_w * dilation_w - dilation_w - pads_left < 0` is not supported
 - `stride_h * (height - 1) - pads_top + 1 < output_h` is not supported
 - `stride_w * (width - 1) - pads_left + 1 < output_w` is not supported
- **group** - `int`:
 - Supports group = 1; when num_input = num_output, also supports group = num_output.
- **dilations** - `int[dilation_h, dilation_w]`:
 - dilation_h range: [1, 32]
 - dilation_w range: [1, 32]
 - When dilation != (1, 1), only kernel = (1, 1) is supported.

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<float16>`):

ConvActivation

Fused Conv and generic Activation computation; activation type is determined by the Activation sub-operator.

FORMULA

```
output = Activation(Conv(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [Conv](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::_convolution
```

ConvRelu

Fused Conv and Relu computation.

FORMULA

```
output = Relu(Conv(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [Conv](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::_convolution
```

ConvClip

Fused Conv and Clip computation.

FORMULA

```
output = Clip(Conv(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [Conv](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::_convolution
```

ConvSigmoid

Fused Conv and Sigmoid computation.

FORMULA

```
output = Sigmoid(Conv(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [Conv](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::_convolution
```

ConvTanh

Fused Conv and Tanh computation.

FORMULA

```
output = Tanh(Conv(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [Conv](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::_convolution
```

ConvElu

Fused Conv and Elu computation.

FORMULA

```
output = Elu(Conv(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [Conv](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::_convolution
```

ConvHardSigmoid

Fused Conv and HardSigmoid computation.

FORMULA

```
output = HardSigmoid(Conv(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [Conv](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::_convolution
```

ConvHardSwish

Fused Conv and HardSwish computation.

FORMULA

```
output = HardSwish(Conv(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [Conv](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::_convolution
```

ConvSwish

Fused Conv and Swish computation.

FORMULA

```
output = Swish(Conv(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [Conv](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::_convolution
```

ConvSoftplus

Fused Conv and Softplus computation.

FORMULA

```
output = Softplus(Conv(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [Conv](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::_convolution
```

ConvMish

Fused Conv and Mish computation.

FORMULA

```
output = Mish(Conv(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [Conv](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::_convolution
```

ConvAdd

Fused Conv and Add computation.

FORMULA

```
output = Add(Conv(input_tensor, weight, bias), residual)
```

INHERITANCE

Inherits from [Conv](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::_convolution
```

EXTRA INPUTS

- **residual** - `T1`:
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

- Second input of Add.
Supports broadcasting; broadcast constraints are the same as Add.

ConvAddRelu

Fused Conv, Add, and Relu computation.

FORMULA

```
output = Relu(Add(Conv(input_tensor, weight, bias), residual))
```

INHERITANCE

Inherits from [Conv](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::_convolution
```

EXTRA INPUTS

- **residual** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - Second input of Add.
Supports broadcasting; broadcast constraints are the same as Add.

ConvPRelu

Fused Conv and PRelu computation.

FORMULA

```
output = PRelu(Conv(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [Conv](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::_convolution
```

ConvTransposeActivation

Fused ConvTranspose and generic Activation computation; activation type is determined by the Activation sub-operator.

FORMULA

```
output = Activation(ConvTranspose(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [ConvTranspose](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::conv_transpose2d
```

ConvTransposeRelu

Fused ConvTranspose and Relu computation.

FORMULA

```
output = Relu(ConvTranspose(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [ConvTranspose](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::conv_transpose2d
```

ConvTransposeClip

Fused ConvTranspose and Clip computation.

FORMULA

```
output = Clip(ConvTranspose(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [ConvTranspose](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::conv_transpose2d
```

ConvTransposeSigmoid

Fused ConvTranspose and Sigmoid computation.

FORMULA

```
output = Sigmoid(ConvTranspose(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [ConvTranspose](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::conv_transpose2d
```

ConvTransposeTanh

Fused ConvTranspose and Tanh computation.

FORMULA

```
output = Tanh(ConvTranspose(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [ConvTranspose](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::conv_transpose2d
```

ConvTransposeElu

Fused ConvTranspose and Elu computation.

FORMULA

```
output = Elu(ConvTranspose(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [ConvTranspose](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::conv_transpose2d
```

ConvTransposeHardSigmoid

Fused ConvTranspose and HardSigmoid computation.

FORMULA

```
output = HardSigmoid(ConvTranspose(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [ConvTranspose](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::conv_transpose2d
```

ConvTransposeHardSwish

Fused ConvTranspose and HardSwish computation.

FORMULA

```
output = HardSwish(ConvTranspose(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [ConvTranspose](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::conv_transpose2d
```

ConvTransposeSwish

Fused ConvTranspose and Swish computation.

FORMULA

```
output = Swish(ConvTranspose(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [ConvTranspose](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::conv_transpose2d
```

ConvTransposeSoftplus

Fused ConvTranspose and Softplus computation.

FORMULA

```
output = Softplus(ConvTranspose(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [ConvTranspose](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::conv_transpose2d
```

ConvTransposeMish

Fused ConvTranspose and Mish computation.

FORMULA

```
output = Mish(ConvTranspose(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [ConvTranspose](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::conv_transpose2d
```

ConvTransposeAdd

Fused ConvTranspose and Add computation.

FORMULA

```
output = Add(ConvTranspose(input_tensor, weight, bias), residual)
```

INHERITANCE

Inherits from [ConvTranspose](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::conv_transpose2d
```

EXTRA INPUTS

- **residual** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

- Second input of Add.
Supports broadcasting; broadcast constraints are the same as Add.

ConvTransposePrelu

Fused ConvTranspose and PRelu computation.

FORMULA

```
output = PRelu(ConvTranspose(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [ConvTranspose](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::conv_transpose2d
```

exConvStreaming

Convolution computation with streaming cache.

FORMULA

```
y = Conv(Concat(cache, x), w, b)
cache = Slice(Concat(cache, Slice(x, width=[0, tc])), width=[end - cache_w,
end])
```

OPERATOR LEVEL

RKNN3 Extension Operator

PYTORCH SOURCE

```
aten::_convolution
```

INPUTS

- **x** - `T1` (required):
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
- **w** - `T1` (required):

- Shape: OIHW[output_channel, input_channel, kernel_h, kernel_w]
- Quantization: per-layer, per-channel
- kernel_h range: [1, 31]
- kernel_w range: [1, 31]
- **b** - `T1` (optional):
 - Shape: C[output_channel]
 - Quantization: per-layer
 - Optional; when there is no bias but tc needs to be passed, b can be an empty input placeholder.
- **tc** - `T2` (optional):
 - Shape: N[1]
 - tc indicates the effective width written to cache this time; the length must be less than the input width.
When there are 3 inputs, if the size of the third input equals the output_channel of weight, it is treated as b; otherwise as tc.

OUTPUTS

- **y** - `T1`:
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

ATTRIBUTES

- **strides** - `int[stride_h, stride_w]`:
 - stride_h range: [1, 7]
 - stride_w range: [1, 7]
- **pads** - `int[pads_top, cache_w, pads_bottom, pads_right]`:
 - pads_top range: [0, 15]
 - pads_bottom range: [0, 15]
 - pads_right range: [0, 15]
 - cache_w is the streaming cache length; must be greater than 0.
- **group** - `int`: —
- **dilations** - `int[dilation_h, dilation_w]`:
 - dilation_h range: [1, 31]
 - dilation_w range: [1, 31]

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<float16>`): **T2** in (`tensor<int32>`):

SPECIAL CASES

Depthwise exConvStreaming When exConvStreaming is used with depthwise convolution group semantics, the following differential specifications apply.

- **Input Differences** — **w**: kernel_h range: [1, 8]; kernel_w range: [1, 8]
- **Attribute Differences** — **dilations**: dilation_h range: [1, 32]; dilation_w range: [1, 32]

NOTES

Subgraph operator; the actual implementation is expanded by createSubGraph at the IR layer; emit only returns success.

Pooling

AveragePool

Performs average pooling on the input tensor with the specified kernel_shape, strides, and pads.

FORMULA

```
Y = average_pool(X, kernel_shape, strides, pads)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::adaptive_avg_pool2d, aten::avg_pool2d
```

INPUTS

- **X** - **T1**:
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - int8/int16 quantization requires height <= 32 and width <= 32

OUTPUTS

- **Y** - **T1**:
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

ATTRIBUTES

- **kernel_shape** - `int[kernel_h, kernel_w]`:
 - kernel_h range: [1, 32]
 - kernel_w range: [1, 32]
- **strides** - `int[stride_h, stride_w]`:
 - stride_h range: [1, 63]
 - stride_w range: [1, 63]
- **pads** - `int[pad_top, pad_left, pad_bottom, pad_right]`:
 - pad_top range: [0, 63]
 - pad_left range: [0, 63]
 - pad_bottom range: [0, 63]
 - pad_right range: [0, 63]
- **auto_pad** - `string`:
 - Only explicit pads are supported.
- **ceil_mode** - `int`:
 - ceil 模式通过补充 bottom/right padding 支持。
- **count_include_pad** - `int`:
 - When set to 1, the mean calculation includes the padding area.
- **dilations** - `int[dilation_h, dilation_w]`:
 - dilation_h range: [1, 1]
 - dilation_w range: [1, 1]

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<float16>`, `tensor<bfloat16>`):

MaxPool

Performs max pooling on the input tensor with the specified kernel_shape, strides, and pads.

FORMULA

```
Y = max_pool(X, kernel_shape, strides, pads)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

`aten::max_pool1d`, `aten::max_pool2d`

INPUTS

- **X - T1**:
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **Y - T1**:
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

ATTRIBUTES

- **kernel_shape - int[kernel_h, kernel_w]**:
 - kernel_h range: [1, 32]
 - kernel_w range: [1, 256]
 - kernel_h * kernel_w <= 256
- **strides - int[stride_h, stride_w]**:
 - stride_h range: [1, 31]
 - stride_w range: [1, 256]
- **pads - int[pad_top, pad_left, pad_bottom, pad_right]**:
 - pad_top range: [0, 63]
 - pad_left range: [0, 63]
 - pad_bottom range: [0, 63]
 - pad_right range: [0, 63]
- **auto_pad - string**:
 - Only explicit pads are supported.
- **ceil_mode - int**:
 - ceil 模式通过补充 bottom/right padding 支持。
- **dilations - int[dilation_h, dilation_w]**:
 - dilation_h range: [1, 1]
 - dilation_w range: [1, 1]

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<float16>`): **T2** in (`tensor<int64>`):

GlobalAveragePool

Performs global average pooling on the H/W spatial dimensions of the input tensor.

FORMULA

```
Y = mean(X, axes=[2, 3])
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::adaptive_avg_pool2d
```

INPUTS

- **X - T1**:
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - int8 quantization requires height <= 32 and width <= 32;

OUTPUTS

- **Y - T1**:
 - Shape: NCHW[batch, channel, 1, 1]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<float16>`, `tensor<bfloat16>`):

GlobalMaxPool

Performs global max pooling on the spatial dimensions of the input tensor, outputting the maximum value per channel.

FORMULA

```
Y[batch, channel, 0, 0] = max(X[batch, channel, :, :])
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::adaptive_max_pool2d
```

INPUTS

- **X** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - Hard constraint: height * width <= 256 and height <= 32.

OUTPUTS

- **Y** - **T1** :
 - Shape: NCHW[batch, channel, 1, 1]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<float16>`):

Elementwise Operations

Add

Performs element-wise binary addition, supporting NumPy-style broadcasting.

FORMULA

```
C = A + B
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::add , aten::add_
```

INPUTS

- **A** - **T1, T2** :
 - Shape: NCHW[batch, channel, height, width]

- Quantization: per-layer
- Supports NumPy-style broadcasting; broadcast constraints are the same as the Expand operator.
- **B** - `T1, T2` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - Supports NumPy-style broadcasting; broadcast constraints are the same as the Expand operator.

OUTPUTS

- **C** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<float16>`): **T2** in (`const_tensor<float16>`):

Sub

Performs element-wise binary subtraction, supporting NumPy-style broadcasting.

FORMULA

`C = A - B`

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

`aten::sub`, `aten::sub_`

INPUTS

- **A** - `T1, T2` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - Supports NumPy-style broadcasting; broadcast constraints are the same as the Expand operator.
- **B** - `T1, T2` :
 - Shape: NCHW[batch, channel, height, width]

- Quantization: per-layer
- Supports NumPy-style broadcasting; broadcast constraints are the same as the Expand operator.

OUTPUTS

- **C** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>` , `tensor<float16>`): **T2** in (`const_tensor<float16>`):

Mul

Performs element-wise binary multiplication, supporting NumPy-style broadcasting.

FORMULA

`C = A * B`

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

`aten::mul` , `aten::mul_`

INPUTS

- **A** - **T1, T2** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - Supports NumPy-style broadcasting; broadcast constraints are the same as the Expand operator.
- **B** - **T1, T2** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - Supports NumPy-style broadcasting; broadcast constraints are the same as the Expand operator.

OUTPUTS

- **C** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<float16>`): **T2** in (`const_tensor<float16>`):

Div

Performs element-wise binary division, supporting NumPy-style broadcasting.

FORMULA

$$C = A / B$$

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

`aten::div`, `aten::true_divide`

INPUTS

- **A** - **T1, T2** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - Supports NumPy-style broadcasting; broadcast constraints are the same as the Expand operator.
- **B** - **T1, T2** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - Supports NumPy-style broadcasting; broadcast constraints are the same as the Expand operator.

OUTPUTS

- **C** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (tensor<float16>): **T2** in (const_tensor<float16>):

Min

Computes the element-wise minimum of multiple input tensors, supporting NumPy-style broadcasting.

FORMULA

```
C = min(A, B, ...)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::min, aten::minimum
```

INPUTS

- **A** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - Supports NumPy-style broadcasting; broadcast constraints are the same as the Expand operator.
- **B** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - Supports NumPy-style broadcasting; broadcast constraints are the same as the Expand operator.

OUTPUTS

- **C** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer, per-channel

DATA TYPE CONSTRAINTS

T1 in (tensor<int8>, tensor<float16>):

Max

Computes the element-wise maximum of multiple input tensors, supporting NumPy-style broadcasting.

FORMULA

```
C = max(A, B, ...)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::max, aten::maximum
```

INPUTS

- **A** - **T1**:
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - Supports NumPy-style broadcasting; broadcast constraints are the same as the Expand operator.
- **B** - **T1**:
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - Supports NumPy-style broadcasting; broadcast constraints are the same as the Expand operator.

OUTPUTS

- **C** - **T1**:
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer, per-channel

DATA TYPE CONSTRAINTS

T1 in (tensor<int8>, tensor<float16>):

Pow

Performs element-wise exponentiation.

FORMULA

```
output = input ^ exponent
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::pow , aten::pow_
```

INPUTS

- **input** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
- **exponent** - `T2` :
 - Shape: S[scalar]

OUTPUTS

- **output** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (tensor<int8> , tensor<float16>): **T2** in (const_tensor<float16>):

Neg

Performs element-wise negation on the input tensor.

FORMULA

```
Y = -X
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::neg , aten::neg_
```

INPUTS

- **X** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **Y** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>` , `tensor<float16>`):

Exp

Computes the element-wise base-e exponential of the input tensor.

FORMULA

```
output = exp(input)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::exp , aten::exp_
```

INPUTS

- **input** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **output** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (tensor<int8> , tensor<float16>):

Sqrt

Performs element-wise square root on the input tensor.

FORMULA

```
output = sqrt(input)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::sqrt , aten::sqrt_
```

INPUTS

- **input** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **output** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (tensor<int8> , tensor<float16>):

Reciprocal

Performs element-wise reciprocal on the input tensor.

FORMULA

```
Y = 1 / X
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::reciprocal, aten::reciprocal_
```

INPUTS

- **X** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **Y** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (tensor<int8>, tensor<float16>):

Floor

Performs element-wise floor (rounds towards negative infinity) on the input tensor.

FORMULA

```
output = floor(input)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::floor, aten::floor_
```

INPUTS

- **input** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **output** - **T1** :
 - Shape: NCHW[batch, channel, height, width]

- Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<float16>`):

Round

Rounding function using round-half-up semantics, formula: $y = \text{Floor}(x + 0.5)$. Differs from ONNX Round's round-half-to-even (banker's rounding).

FORMULA

`y = Floor(x + 0.5)`

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

`aten::round`, `aten::round_`

INPUTS

- **x** - `T1`:
 - Shape: NCHW[batch, channel, height, width]

OUTPUTS

- **y** - `T1`:
 - Shape: NCHW[batch, channel, height, width]

DATA TYPE CONSTRAINTS

T1 in (`tensor<float16>`):

Sin

Computes the element-wise sine function.

FORMULA

`y = sin(x)`

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::sin , aten::sin_
```

INPUTS

- **input** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **output** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>` , `tensor<float16>`):

NOTES

Implementation first applies $\text{mod } 2\pi$ to the input, then performs a LUT lookup.

Cos

Computes the element-wise cosine function.

FORMULA

```
y = cos(x)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::cos , aten::cos_
```

INPUTS

- **input** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **output** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<float16>`):

NOTES

Implementation first applies $\text{mod } 2\pi$ to the input, then performs a LUT lookup.

exFastExp

Computes the element-wise natural exponential using an approximation for improved performance.

FORMULA

`output  $\approx$  exp(input)`

OPERATOR LEVEL

RKNN3 Extension Operator

PYTORCH SOURCE

`aten::exp`, `aten::exp_`

INPUTS

- **input** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **output** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<float16>`):

exTwoPow

Computes the element-wise power of 2.

FORMULA

$$y = 2^x$$

OPERATOR LEVEL

RKNN3 Extension Operator

PYTORCH SOURCE

```
aten::pow
```

INPUTS

- **input** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **output** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<float16>`):

AddRelu

Fused Add and Relu computation.

FORMULA

$$C = \text{Relu}(A + B)$$

INHERITANCE

Inherits from [Add](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::add , aten::add_
```

MulRelu

Fused Mul and Relu computation.

FORMULA

```
C = Relu(A * B)
```

INHERITANCE

Inherits from [Mul](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::mul , aten::mul_
```

Activation Functions

Relu

Rectified Linear Unit, performs $\max(\text{input}, 0)$ element-wise.

FORMULA

```
output = max(input, 0)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::relu , aten::relu6 , aten::relu_
```

INPUTS

- **input** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **output** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>` , `tensor<float16>`):

Sigmoid

Applies the Sigmoid function element-wise to the given input tensor.

FORMULA

$$y = 1 / (1 + \exp(-x))$$

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

`aten::sigmoid` , `aten::sigmoid_`

INPUTS

- **input** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **output** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>` , `tensor<float16>`):

Clip

Clips the input tensor elements to the [min, max] range; out-of-bounds values are clamped to the boundary.

FORMULA

```
output = Clip(input, min, max)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::clamp, aten::clamp_
```

INPUTS

- **input** - `T1` (required):
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
- **min** - `T2` (optional):
 - Shape: S[scalar]
 - Clip lower bound, scalar. Omitted means no lower bound.
- **max** - `T2` (optional):
 - Shape: S[scalar]
 - Clip upper bound, scalar. Omitted means no upper bound.

OUTPUTS

- **output** - `T1`:
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<float16>`): **T2** in (`const_tensor<float16>`):

LeakyRelu

Leaky Rectified Linear Unit; positive values remain unchanged, negative values are scaled by a fixed ratio alpha.

FORMULA

```
output = input if input >= 0 else alpha * input
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::leaky_relu, aten::leaky_relu_
```

INPUTS

- **input** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **output** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

ATTRIBUTES

- **alpha** - `float[alpha]`:
 - alpha range: [0, 1]
 - Negative value scaling factor.

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<float16>`):

PRelu

Parametric Rectified Linear Unit; positive values remain unchanged, negative values are scaled by a learnable parameter slope.

FORMULA

```
output = input if input >= 0 else slope * input
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::prelu, aten::prelu_
```

INPUTS

- **input** - `T1` :

- Shape: NCHW[batch, channel, height, width]
- Quantization: per-layer, per-channel
- **slope** - `T2` :
 - Shape: C[channel]
 - Quantization: per-channel
 - Supports a single scalar or C-dimension coefficients.

OUTPUTS

- **output** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer, per-channel

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<float16>`): **T2** in (`const_tensor<float16>`):

Elu

Applies the Elu function element-wise, alpha = 1.

FORMULA

$f(x) = \alpha * (\exp(x) - 1) \text{ for } x < 0; f(x) = x \text{ for } x \geq 0$

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

`aten::elu`, `aten::elu_`

INPUTS

- **input** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **output** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (tensor<int8> , tensor<float16>):

Softplus

Applies the Softplus function element-wise.

FORMULA

$$y = \ln(\exp(x) + 1)$$

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

`aten::softplus`

INPUTS

- **X** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **Y** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (tensor<int8> , tensor<float16>):

HardSigmoid

Applies the HardSigmoid function element-wise, controlled by alpha and beta attributes.

FORMULA

$$y = \max(0, \min(1, \alpha * x + \beta))$$

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::hardsigmoid, aten::hardsigmoid_
```

INPUTS

- **input** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **output** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

ATTRIBUTES

- **alpha** - **float**:
 - Default 0.2 (consistent with the ONNX standard).
- **beta** - **float**:
 - Default 0.5.

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<float16>`):

HardSwish

Applies the HardSwish function element-wise, $\alpha = 1/6$, $\beta = 0.5$.

FORMULA

```
y = x * max(0, min(1, alpha * x + beta))
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::hardswish, aten::hardswish_
```

INPUTS

- **X** - `T1` :

- Shape: NCHW[batch, channel, height, width]
- Quantization: per-layer

OUTPUTS

- **Y** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (tensor<int8> , tensor<float16>):

Tanh

Computes the element-wise hyperbolic tangent of the given input tensor.

FORMULA

$$y = \tanh(x)$$

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

aten::tanh , aten::tanh_

INPUTS

- **X** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **Y** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (tensor<int8> , tensor<float16>):

Mish

Applies the Mish function element-wise.

FORMULA

$$y = x * \tanh(\text{softplus}(x)) = x * \tanh(\ln(1 + \exp(x)))$$

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::mish
```

INPUTS

- **input** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **output** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<float16>`):

exGelu

Applies the Gelu function element-wise.

FORMULA

$$y = x * \Phi(x), \text{ when } \Phi \text{ uses } \tanh \text{ approximation: } y = 0.5 * x * (1 + \tanh(\sqrt{2/\pi} * (x + 0.044715 * x^3)))$$

OPERATOR LEVEL

RKNN3 Extension Operator

PYTORCH SOURCE

```
aten::gelu
```

INPUTS

- **input** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **output** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>` , `tensor<float16>`):

exHardSwish

Applies the HardSwish function element-wise, $\alpha = 1/6$, $\beta = 0.5$.

FORMULA

```
y = x * max(0, min(1, alpha * x + beta))
```

OPERATOR LEVEL

RKNN3 Extension Operator

PYTORCH SOURCE

```
aten::hardswish , aten::hardswish_
```

INPUTS

- **X** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **Y** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (tensor<int8> , tensor<float16>):

exMish

Applies the Mish function element-wise.

FORMULA

$$y = x * \tanh(\text{softplus}(x)) = x * \tanh(\ln(1 + \exp(x)))$$

OPERATOR LEVEL

RKNN3 Extension Operator

PYTORCH SOURCE

```
aten::mish , aten::mish_
```

INPUTS

- **input** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **output** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (tensor<int8> , tensor<float16>):

exSwish

Applies the Swish function (also known as SiLU) element-wise.

FORMULA

$$y = x * \text{sigmoid}(x)$$

OPERATOR LEVEL

RKNN3 Extension Operator

PYTORCH SOURCE

```
aten::silu, aten::silu_
```

INPUTS

- **X** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **Y** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (tensor<int8>, tensor<float16>):

exGlu

Splits the input tensor into two equal parts along the specified axis, applies sigmoid gating to the second part, and multiplies it with the first part.

FORMULA

```
a, b = np.split(X, 2, axis=axis)
Y = a * sigmoid(b)
```

OPERATOR LEVEL

RKNN3 Extension Operator

PYTORCH SOURCE

```
aten::glu
```

INPUTS

- **X** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - The length of the dimension specified by axis must be even.
The NPU subgraph implementation primarily targets axis = 1 channel dimension

splitting.

OUTPUTS

- **Y - T1 :**
 - Shape: NCHW[batch, out_channel, height, width]
 - Quantization: per-layer
 - The output shape is consistent with the input shape, but the dimension specified by axis becomes half of the input.
When axis = 1, out_channel = channel / 2.

ATTRIBUTES

- **axis - int:**
 - Specifies the dimension for GLU split; valid range is [-rank, rank - 1].
Negative axes are converted to positive axes following NumPy semantics.

DATA TYPE CONSTRAINTS

T1 in (tensor<int8>, tensor<float16>):

exLogit

Applies the logit activation function to the input tensor.

FORMULA

$$Y = \log(X / (1 - X))$$

OPERATOR LEVEL

RKNN3 Extension Operator

PYTORCH SOURCE

```
aten::logit
```

INPUTS

- **X - T1 :**
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **Y - T1 :**

- Shape: NCHW[batch, channel, height, width]
- Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (tensor<int8> , tensor<float16>):

Comparison & Logical Operations

Equal

Compares whether two tensors are equal element-wise, outputs a bool result, supporting NumPy-style broadcasting.

FORMULA

$C = (A == B)$

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

aten::eq , aten::eq_

INPUTS

- **A - T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
- **B - T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **C - T2** :
 - Shape: NCHW[batch, channel, height, width]
 - Boolean semantic result; RKNN stores it in int8 memory layout.

DATA TYPE CONSTRAINTS

T1 in (tensor<int8> , tensor<float16>) : **T2** in (tensor<bool> , tensor<int8>):

Greater

Compares whether A is greater than B element-wise, outputs a bool result, supporting NumPy-style broadcasting.

FORMULA

$$C = (A > B)$$

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

`aten::gt`, `aten::gt_`

INPUTS

- **A** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - Supports NumPy-style broadcasting.
- **B** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - Supports NumPy-style broadcasting.

OUTPUTS

- **C** - `T2` :
 - Shape: NCHW[batch, channel, height, width]

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<float16>`): **T2** in (`tensor<bool>`, `tensor<int8>`):

GreaterOrEqual

Compares whether A is greater than or equal to B element-wise, outputs a bool result, supporting NumPy-style broadcasting.

FORMULA

$$C = (A \geq B)$$

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

`aten::ge` , `aten::ge_`

INPUTS

- **A** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
- **B** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **C** - `T2` :
 - Shape: NCHW[batch, channel, height, width]
 - Boolean semantic result; RKNN stores it in int8 memory layout.

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>` , `tensor<float16>`): **T2** in (`tensor<bool>` , `tensor<int8>`):

Less

Compares whether A is less than B element-wise, outputs a bool result, supporting NumPy-style broadcasting.

FORMULA

`C = (A < B)`

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

`aten::lt` , `aten::lt_`

INPUTS

- **A** - `T1` :

- Shape: NCHW[batch, channel, height, width]
- Quantization: per-layer
- Supports NumPy-style broadcasting.
- **B** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - Supports NumPy-style broadcasting.

OUTPUTS

- **C** - **T2** :
 - Shape: NCHW[batch, channel, height, width]

DATA TYPE CONSTRAINTS

T1 in (tensor<int8> , tensor<float16>): **T2** in (tensor<bool> , tensor<int8>):

LessOrEqual

Compares whether A is less than or equal to B element-wise, outputs a bool result, supporting NumPy-style broadcasting.

FORMULA

$C = (A \leq B)$

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

aten::le , aten::le_

INPUTS

- **A** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - Supports NumPy-style broadcasting.
- **B** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

- Supports NumPy-style broadcasting.

OUTPUTS

- **C** - **T2** :
 - Shape: NCHW[batch, channel, height, width]

DATA TYPE CONSTRAINTS

T1 in (tensor<int8> , tensor<float16>): **T2** in (tensor<bool> , tensor<int8>):

Not

Performs element-wise logical NOT on the input tensor.

FORMULA

$$Y = !X$$

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

aten::logical_not

INPUTS

- **X** - **T1** :
 - Shape: NCHW[batch, channel, height, width]

OUTPUTS

- **Y** - **T1** :
 - Shape: NCHW[batch, channel, height, width]

DATA TYPE CONSTRAINTS

T1 in (tensor<bool>):

And

Performs element-wise logical AND, outputs a bool result, supporting NumPy-style broadcasting.

FORMULA

```
C = A && B
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::bitwise_and, aten::logical_and
```

INPUTS

- **A** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Supports NumPy-style broadcasting; broadcast constraints are the same as the Expand operator.
- **B** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Supports NumPy-style broadcasting; broadcast constraints are the same as the Expand operator.

OUTPUTS

- **C** - `T1` :
 - Shape: NCHW[batch, channel, height, width]

DATA TYPE CONSTRAINTS

T1 in (`tensor<bool>`, `tensor<int8>`):

Where

Selects elements from x or y based on the condition, supporting NumPy-style broadcasting.

FORMULA

```
output = select(condition, x, y)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::where
```

INPUTS

- **condition** - T1 :
 - Shape: NCHW[batch, channel, height, width]
- **x** - T2 :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - Supports NumPy-style broadcasting.
- **y** - T2 :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - Supports NumPy-style broadcasting.

OUTPUTS

- **output** - T2 :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (tensor<bool>): **T2** in (tensor<int8> , tensor<float16>):

Reduction Operations

ReduceMax

Computes the maximum of the input tensor along the specified axis, supporting multiple axes and keepdims.

FORMULA

```
output = ReduceMax(input, axes, keepdims)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::amax , aten::max
```

INPUTS

- **input** - T1 :

- Shape: NCHW[batch, channel, height, width]
- Quantization: per-layer

OUTPUTS

- **output** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - When keepdims=1, the reduced axis has size 1; when keepdims=0, the axis is removed from the output. The actual output rank depends on axes and keepdims.

ATTRIBUTES

- **axes** - `int[]`:
 - Axis for computing the maximum value.
- **keepdims** - `int`:
 - Whether to keep the reduced dimensions (1 keeps, 0 removes).

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<float16>`):

ReduceMean

Reduces the input tensor by computing the mean along the specified axes.

FORMULA

```
Y = mean(data, axes)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::mean
```

INPUTS

- **data** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - For int8/int16 quantization with axes = [2], [3], [2, 3], or [1, 2, 3], height <= 32 and

width ≤ 32 are required

- **axes** - `T2` :
 - Shape: Axes[num_axes]
 - Specifies the dimensions to reduce; negative axes are converted to positive axes following NumPy semantics.
Supports axes = [1], [2], [3], [2, 3], [1, 2, 3]

OUTPUTS

- **reduced** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<float16>`, `tensor<bfloat16>`): **T2** in (`const_tensor<int64>`):

ReduceSum

Reduces the input tensor by computing the sum along the specified axes.

FORMULA

```
Y = sum(data, axes)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::sum
```

INPUTS

- **data** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - For int8/int16 quantization with axes = [2], [3], [2, 3], or [1, 2, 3], height ≤ 32 and width ≤ 32 are required
- **axes** - `T2` :
 - Shape: Axes[num_axes]

- Specifies the dimensions to reduce; negative axes are converted to positive axes following NumPy semantics.
Supports axes = [1], [2], [3], [2, 3], [1, 2, 3]

OUTPUTS

- **reduced** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (tensor<int8> , tensor<float16> , tensor<bfloat16>): **T2** in (const_tensor<int64>):

ArgMax

Returns the indices of the maximum values along the specified axis.

FORMULA

```
output = ArgMax(input, axis)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::argmax
```

INPUTS

- **input** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **output** - **T2** :
 - Shape: NCHW[batch, channel, height, width]

ATTRIBUTES

- **axis** - **int**:
 - Specifies the axis for argmax operation; valid range [0, ndim-1].

- **keepdims - int:**
 - Whether to keep the output dimensions the same as the input. 0 means no, 1 means yes.
- **select_last_index - int:**
 - When multiple maximum values exist, whether to select the last index. 0 selects the first, 1 selects the last.

DATA TYPE CONSTRAINTS

T1 in (tensor<int8> , tensor<float16>): **T2** in (tensor<int32>):

ArgMin

Returns the indices of the minimum values along the specified axis.

FORMULA

```
output = ArgMin(input, axis)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::argmin
```

INPUTS

- **input - T1 :**
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **output - T2 :**
 - Shape: NCHW[batch, channel, height, width]

ATTRIBUTES

- **axis - int:**
 - Specifies the axis for argmin operation; valid range [0, ndim-1].
- **keepdims - int:**
 - Whether to keep the output dimensions the same as the input. 0 means no, 1 means yes.

- **select_last_index - int:**

- When multiple minimum values exist, whether to select the last index. 0 selects the first, 1 selects the last.

DATA TYPE CONSTRAINTS

T1 in (tensor<int8>, tensor<float16>): **T2** in (tensor<int32>):

TopK

Retrieves the K largest (or smallest, determined by largest) values and their indices along the specified axis.

FORMULA

(values, indices) = TopK(X, K, axis, largest, sorted)

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

aten::topk

INPUTS

- **X** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
- **K** - **T3** :
 - Shape: S[scalar]
 - The top K values, scalar, stored in an int64 tensor.

OUTPUTS

- **values** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - The top K values.
- **indices** - **T2** :
 - Shape: NCHW[batch, channel, height, width]
 - Index of the corresponding value. The ONNX standard is int64; RKNN actually outputs int32 or int64.

ATTRIBUTES

- **axis - int:**
 - Axis for computation; default -1.
- **largest - int:**
 - 1 selects maximum, 0 selects minimum, default 1.
- **sorted - int:**
 - 1 outputs in descending order, 0 does not guarantee order, default 1.

DATA TYPE CONSTRAINTS

T1 in (tensor<int8> , tensor<float16>): **T2** in (tensor<int32> , tensor<int64>): **T3** in (tensor<int64>):

Tensor Transform

Reshape

Returns a tensor with the specified shape without changing the input data or element count.

FORMULA

```
output = reshape(input, shape)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::reshape , aten::reshape_
```

INPUTS

- **input - T1 :**
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **output - T1 :**
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

ATTRIBUTES

- **shape - int:**
 - Specifies the output tensor shape.

DATA TYPE CONSTRAINTS

T1 in (tensor<int8> , tensor<float16>):

Transpose

Transposes the input tensor according to the axis order specified by perm.

FORMULA

```
Y = transpose(X, perm)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::permute , aten::transpose , aten::transpose_
```

INPUTS

- **X - T1, T2 :**
 - Shape: NCHW[n1, c1, h1, w1]
 - Quantization: per-layer
 - For detailed constraint rules, see the perm attribute list.

OUTPUTS

- **Y - T1, T2 :**
 - Shape: NCHW[n2, c2, h2, w2]
 - Quantization: per-layer
 - For detailed constraint rules, see the perm attribute list.

ATTRIBUTES

- **perm - int[d0, d1, d2, d3]:**
 - Axis order for transpose. Constraints for different perm parameters are listed in the perm constraint table below.

DATA TYPE CONSTRAINTS

T1 in (tensor<int8>): **T2** in (tensor<int16> , tensor<float16>):

NOTES

Different perm parameter constraints are as follows (where p = 268435456, q = 65536):

perm	T1 Constraint	T2 Constraint
[0,1,2,3]	No restriction	No restriction
[0,1,3,2]	$h1\%16=0, h1w1<16p, h1w1<2q$	$h1\%8=0, h1w1<8p, h1w1<2q$
[0,2,3,1]	$c1<8192, h1w1\%8=0, w1c1<p*16$	$c1<8192, h1w1\%8=0, w1c1<p*8$
[0,2,1,3]	$h1\%16=0, h1w1<16p, h1w1<2q, c1*w1<p$	$h1\%8=0, h1w1<8p, h1w1<2q, c1*w1<p$
[0,3,1,2]	$h1w1<p, w1<8192, h1c1<p$	$h1w1<p, w1<8192, h1c1<p$
[0,3,2,1]	$h1\%16=0, h1w1<16p, h1w1<2q, c1<p*16$	$h1\%8=0, h1w1<8p, h1w1<2q, c1<p*8$
[1,0,2,3]	$h1w1<16p, n1<8192, h1*w1<p$	$h1w1<8p, n1<8192, h1*w1<p$
[1,0,3,2]	$h1\%16=0, c1h1w1<16p, h1w1<2*q$	$h1\%8=0, c1h1w1<8p, h1w1<2*q$
[1,2,0,3]	$n1w1<p16, c1h1w1<p, w1\%16=0$	$n1w1<p8, c1h1w1<p, w1\%8=0$
[1,2,3,0]	$h1w1<p, n1<p16, c1h1w1<p16, n1w1<p*16$	$h1w1<p, n1<p8, c1h1w1<p8, n1w1<p*8$
[1,3,0,2]	$c1w1<8192, n1h1<p, w1\%16=0$	$c1w1<8192, n1h1<p, w1\%8=0$
[1,3,2,0]	$c1h1w1<p, n1*h1<p, w1<8192, w1\%16=0$	$c1h1w1<p, n1*h1<p, w1<8192, w1\%8=0$
[2,1,0,3]	$c1<8192, c1h1w1<p, n1*w1<p$	$c1<8192, c1h1w1<p, n1*w1<p$
[2,1,3,0]	$c1h1w1<p, n1w1<p16, c1<8192, n1*w1\%4=0$	$c1h1w1<p, n1w1<p8, c1<8192, n1*w1\%4=0$
[2,0,3,1]	$c1<8192, c1h1w1<p$	$c1<8192, c1h1w1<p$
[2,0,1,3]	$n1c1h1w1<p, h1w1\%16=0$	$n1c1h1w1<p, h1w1\%8=0$
[2,3,1,0]	$c1h1w1<p, n1c1<p, h1w1<8192, h1*w1\%16=0$	$c1h1w1<p, n1c1<p, h1w1<8192, h1*w1\%8=0$
[2,3,0,1]	$h1w1<8192, n1c1<p$	$h1w1<8192, n1c1<p$
[3,0,2,1]	$h1w1<p, n1c1<8192, c1h1<p, n1w1<8192, n1*w1\%16=0$	$h1w1<p, n1c1<8192, c1h1<p, n1w1<8192, n1*w1\%8=0$
[3,0,1,2]	$h1w1<p, n1c1<8192, c1h1<p, n1w1<8192, n1h1w1\%16=0$	$h1w1<p, n1c1<8192, c1h1<p, n1w1<8192, n1h1w1\%8=0$
[3,2,0,1]	$h1w1<p, n1c1<8192, n1c1w1<p$	$h1w1<p, n1c1<8192, n1c1w1<p$
[3,2,1,0]	$c1h1w1<p, h1w1<8192, 2h1w1<q, n1c1<p*16, h1\%16=0$	$c1h1w1<p, h1w1<8192, 2h1w1<q, n1c1<p*8, h1\%8=0$
[3,1,0,2]	$h1w1<p, c1w1<8192, n1*h1<p$	$h1w1<p, c1w1<8192, n1*h1<p$
[3,1,2,0]	$h1w1<p, n1c1<8192, n1h1<p, n1c1*w1<p$	$h1w1<p, n1c1<8192, n1h1<p, n1c1*w1<p$

Concat

Concatenates multiple tensors along the specified axis into a single tensor; all inputs must have the same size except along the axis dimension.

FORMULA

```
output = Concat(input_tensor_list, axis)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::cat
```

INPUTS

- **input_tensor** - **T**:
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - Input is a list of tensors; all tensors must have the same size except along the axis dimension.

OUTPUTS

- **output** - **T**:
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

ATTRIBUTES

- **axis** - **int[axis]**:
 - axis range: [-4, 3]
 - Concatenation axis; RKNN requires 4D NCHW input, supports negative axes.

DATA TYPE CONSTRAINTS

T in (**tensor<int8>**, **tensor<float16>**):

Split

Splits the input tensor into multiple output tensors along the specified axis.

FORMULA

```
outputs = Split(input_tensor, axis, num_outputs, split)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::split
```

INPUTS

- **input_tensor** - **T**:
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **output** - **T**:
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - One or more output tensors; the count is determined by num_outputs, and the shape depends on the split.

ATTRIBUTES

- **axis** - **int[axis]**:
 - axis range: [-4, 3]
 - Split axis; RKNN requires 4D NCHW input, supports negative axes.
- **num_outputs** - **int**: —
- **split** - **int[]**:
 - Specifies the size of each output along the axis dimension after splitting.

DATA TYPE CONSTRAINTS

T in (`tensor<int8>`, `tensor<float16>`):

Expand

Broadcasts the input tensor to the target shape.

FORMULA

```
output = Expand(input, target_shape)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::expand , aten::expand_as
```

INPUTS

- **input** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **output** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>` , `tensor<float16>`):

Slice

Obtains a slice of the input tensor following NumPy slicing rules.

FORMULA

```
output = Slice(data, starts, ends, axes, steps)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::narrow , aten::slice
```

INPUTS

- **data** - `T` (required):
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
- **starts** - `T2` (required):

- Shape: D[dim]
- Per-axis split start positions.
- **ends** - T2 (required):
 - Shape: D[dim]
 - Per-axis split end positions.
- **axes** - T2 (optional):
 - Shape: D[dim]
 - Selects the split axis; supports axes 0~3, supports multiple axes simultaneously; when omitted, all axes are processed.
- **steps** - T2 (optional):
 - Shape: D[dim]
 - Per-axis split step sizes. When height*width==1 and channel%subc==0 and starts[1]%subc==0 and ends[1]%subc==0 and steps[1]<=subc, steps can be non-1; otherwise, the NPU can only run when steps==[1,1,1,1].

OUTPUTS

- **output** - T:
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T in (tensor<int8>, tensor<float16>): T2 in (tensor<int64>):

Pad

Pads the input tensor with the specified pads and mode.

FORMULA

```
output = Pad(data, pads, mode, constant_value)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::constant_pad_nd
```

INPUTS

- **data** - `T1` (required):
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
- **pads** - `T2` (required):
 - Shape: PADS[batch_begin, channel_begin, height_begin, width_begin, batch_end, channel_end, height_end, width_end]
- **constant_value** - `T1` (optional):
 - Shape: S[scalar]
 - Quantization: per-layer
 - Same type as data; only effective in constant mode, defaults to 0.

OUTPUTS

- **output** - `T1`:
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

ATTRIBUTES

- **mode** - **string**:
 - constant mode has no range restrictions; edge mode only supports h/w directions.

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>` , `tensor<int16>` , `tensor<float16>` , `tensor<bfloat16>`): **T2** in (`tensor<int64>`):

SPECIAL CASES

Reflect Pad In reflect mode, input shape components have additional range restrictions.

- **Input Differences** — **data**: channel range: [1, 8192]; height range: [1, 8192]; width range: [1, 8176]

DepthToSpace

Rearranges the channel dimension of the input tensor back to H/W spatial dimensions by blocksize.

FORMULA

```
N, C, H, W = X.shape
b = blocksize
out_channel = C // (b * b)
Y = np.empty((N, out_channel, H * b, W * b), dtype=X.dtype)

for n in range(N):
    for c in range(out_channel):
        for h in range(H * b):
            for w in range(W * b):
                offset = (h % b) * b + (w % b)
                in_c = c + offset * out_channel
                Y[n, c, h, w] = X[n, in_c, h // b, w // b]
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::pixel_shuffle
```

INPUTS

- **input** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - channel must be divisible by blocksize * blocksize.

OUTPUTS

- **output** - `T1` :
 - Shape: NCHW[batch, out_channel, out_height, out_width]
 - Quantization: per-layer
 - out_channel = channel / (blocksize * blocksize).
 - out_height = height * blocksize.
 - out_width = width * blocksize.

ATTRIBUTES

- **blocksize** - `int`:
 - Spatial block size; must be greater than 0.

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<int16>`, `tensor<float16>`):

SpaceToDepth

Rearranges H/W spatial blocks of the input tensor to the channel dimension.

FORMULA

```
N, C, H, W = X.shape
b = blocksize
Y = np.empty((N, C * b * b, H // b, W // b), dtype=X.dtype)

for n in range(N):
    for c in range(C):
        for h in range(H):
            for w in range(W):
                out_c = c + ((h % b) * b + (w % b)) * C
                Y[n, out_c, h // b, w // b] = X[n, c, h, w]
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::pixel_unshuffle
```

INPUTS

- **input** - **T1**:
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - height and width must be divisible by blocksize.

OUTPUTS

- **output** - **T1**:
 - Shape: NCHW[batch, out_channel, out_height, out_width]
 - Quantization: per-layer
 - out_channel = channel * blocksize * blocksize.
out_height = height / blocksize.
out_width = width / blocksize.

ATTRIBUTES

- **blocksize** - **int**:
 - Spatial block size; must be greater than 0.

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>` , `tensor<int16>` , `tensor<float16>`):

Resize

Resizes the spatial dimensions of the input tensor according to the specified scales or sizes, supporting nearest and linear interpolation modes.

FORMULA

```
Y = Resize(X, scales, mode, coordinate_transformation_mode, nearest_mode)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::upsample_bilinear2d , aten::upsample_nearest2d
```

INPUTS

- **X** - `T1` (required):
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
- **roi** - `T3` (optional):
 - Shape: N[8]
 - Crop region coordinates [start_x, start_y, start_z, start_t, end_x, end_y, end_z, end_t]; only the default value [0,0,0,0,1,1,1,1] is supported.
- **scales** - `T3` (optional):
 - Shape: N[4]
 - Per-dimension scale factors [batch_scale, channel_scale, height_scale, width_scale], mutually exclusive with sizes. In linear and nearest modes, only integer scaling is supported, and batch/channel dimension scale must be 1, H/W dimension scale ≤ 8 .
- **sizes** - `T4` (optional):
 - Shape: N[4]
 - Target output size [batch, channel, height, width], mutually exclusive with scales. In linear and nearest modes, only integer scaling is supported, batch/channel dimensions remain unchanged, and H/W dimension multiplier ≤ 8 .

OUTPUTS

- **Y - T1 :**
 - Shape: NCHW[batch, channel, out_height, out_width]
 - Quantization: per-layer

ATTRIBUTES

- **mode - string:**
 - linear and nearest modes are internally implemented via convolution simulation.
- **coordinate_transformation_mode - string:**
 - tf_crop_and_resize is not supported.
- **cubic_coeff_a - float:**
 - Cubic mode coefficient; currently only supports the default value -0.75.
- **exclude_outside - int:**
 - Excluding outer points is not supported.
- **extrapolation_value - float:**
 - Extrapolation value setting is not supported.
- **nearest_mode - string:**
 - Rounding method in nearest mode.
- **antialias - int:**
 - Antialiasing is not supported.
- **keep_aspect_ratio_policy - string:**
 - Only the default stretch mode is supported.
- **axes - int[axis_0, axis_1, axis_2, axis_3]:**
 - Specifies the scaling dimension; the current implementation does not impose additional constraints on this attribute.

DATA TYPE CONSTRAINTS

T1 in (tensor<int8> , tensor<float16>): **T3** in (tensor<float32>): **T4** in (tensor<int64>):

exWindow

exWindow is derived from the window_partition and window_reverse operations in https://github.com/microsoft/Swin-Transformer/blob/main/models/swin_transformer.py . In the original paper, the input feature is divided into multiple non-overlapping windows, each of size window_size × window_size. Both input and output features must be 4D shapes, and division is performed along both height and width dimensions.

FORMULA

```
B, C, H, W = X.shape

if mode == "partition":
    x = X.reshape(B, C, num_windows[0], window_sizes[0],
                  num_windows[1], window_sizes[1])
    y = np.transpose(x, (0, 1, 2, 4, 3, 5))
    Y = y.reshape(B, C,
                  num_windows[0] * num_windows[1],
                  window_sizes[0] * window_sizes[1])

elif mode == "reverse":
    x = X.reshape(B, C, num_windows[0], num_windows[1],
                  window_sizes[0], window_sizes[1])
    y = np.transpose(x, (0, 1, 2, 4, 3, 5))
    Y = y.reshape(B, C,
                  num_windows[0] * window_sizes[0],
                  num_windows[1] * window_sizes[1])

elif mode == "partition_num_first":
    x = X.reshape(B, C, window_sizes[0], num_windows[0],
                  window_sizes[1], num_windows[1])
    y = np.transpose(x, (0, 1, 3, 5, 2, 4))
    Y = y.reshape(B, C,
                  num_windows[0] * num_windows[1],
                  window_sizes[0] * window_sizes[1])
```

OPERATOR LEVEL

RKNN3 Extension Operator

INPUTS

- **X - T1**:
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - partition requires height = num_h * win_h, width = num_w * win_w.
reverse requires height = num_h * num_w, width = win_h * win_w.
partition_num_first requires height = win_h * num_h, width = win_w * num_w.

OUTPUTS

- **Y - T1**:
 - Shape: NCHW[batch, channel, out_height, out_width]
 - Quantization: per-layer
 - partition and partition_num_first output out_height = num_h * num_w, out_width = win_h * win_w.
reverse outputs out_height = num_h * win_h, out_width = num_w * win_w.

ATTRIBUTES

- **mode - string:**
 - partition follows the window_partition implementation.
reverse follows the window_reverse implementation.
The difference between partition_num_first and partition is that partition_num_first divides H/W by num_windows, while partition divides H/W by window_sizes.
- **num_windows - int[num_h, num_w]:**
 - num_h range: [1, 8192]
 - num_w range: [1, 8192]
 - 2D vector, number of windows. Defines how many windows the height and width of the input feature are divided into.
- **window_sizes - int[win_h, win_w]:**
 - win_h range: [1, 8192]
 - win_w range: [1, 8192]
 - 2D vector, window size. Defines the height and width of each window. Attention computation and feature extraction within the window are based on this size.

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<float16>`):

Index & Scatter

Gather

Gathers elements from the input tensor along the specified axis based on indices.

FORMULA

```
output = Gather(input, indices, axis)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::embedding, aten::gather
```

INPUTS

- **input - T1 :**
 - Shape: NCHW[batch, channel, height, width]

- Quantization: per-layer
- **indices** - `T2` :
 - Shape: INDICES[batch, index]

OUTPUTS

- **output** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

ATTRIBUTES

- **axis** - `int`:
 - Gathers elements along this axis.

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>` , `tensor<float16>`): **T2** in (`tensor<int32>`):

GatherElements

Gathers elements from the input tensor along the specified axis by indices; the output shape matches that of indices.

FORMULA

```
output = GatherElements(data, indices, axis)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::gather
```

INPUTS

- **data** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
- **indices** - `T2` :
 - Shape: NCHW[batch, channel, height, width]
 - Supports both constant and non-constant indices.

OUTPUTS

- **output** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

ATTRIBUTES

- **axis** - `int`:
 - Gathers elements along this axis; negative values are normalized to the valid range.

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<float16>`, `tensor<int32>`): **T2** in (`tensor<int32>`):

GatherND

Gathers slices from the input tensor based on indices and combines them into a new output tensor.

FORMULA

```
output = GatherND(data, indices, batch_dims)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::gather_nd
```

INPUTS

- **data** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
- **indices** - `T2` :
 - Shape: NCHW[batch, channel, height, width]

OUTPUTS

- **output** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

ATTRIBUTES

- **batch_dims - int:**
 - Specifies how many leading dimensions are treated as batch dimensions; no gather operation is performed on these dimensions.

DATA TYPE CONSTRAINTS

T1 in (tensor<int8> , tensor<float16>): **T2** in (tensor<int32>):

ScatterND

Scatters updates into the data tensor at positions specified by indices to produce an output tensor, supporting reduction modes.

FORMULA

```
output = ScatterND(data, indices, updates, reduction)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::scatter_nd
```

INPUTS

- **data - T1 :**
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
- **indices - T2 :**
 - Shape: INDICES[num_index, index_dim]
- **updates - T1 :**
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **output - T1 :**
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

ATTRIBUTES

- **reduction - string:**
 - Specifies the reduction mode for scatter.

DATA TYPE CONSTRAINTS

T1 in (tensor<int8> , tensor<float16>): **T2** in (tensor<int32> , tensor<int64>):

Normalization

BatchNormalization

Performs batch normalization on the input tensor (ref: <https://arxiv.org/abs/1502.03167>).

FORMULA

```
output = (input - mean) / sqrt(variance + epsilon)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::batch_norm
```

INPUTS

- **X** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
- **scale** - **T2** :
 - Shape: C[channel]
 - Quantization: per-channel
- **B** - **T2** :
 - Shape: C[channel]
 - Quantization: per-channel
- **mean** - **T2** :
 - Shape: C[channel]
 - Quantization: per-channel
 - During inference, it is folded into scale/B together with var and does not participate in computation at runtime.
- **var** - **T2** :

- Shape: C[channel]
- Quantization: per-channel
- During inference, it is folded into scale/B together with mean and does not participate in computation at runtime.

OUTPUTS

- **output** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

ATTRIBUTES

- **epsilon** - float:
 - A small real number added to the standard deviation to prevent division by zero; default 1e-5.

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>` , `tensor<float16>`): **T2** in (`const_tensor<float16>`):

BatchNormRelu

Fused BatchNormalization and Relu computation.

FORMULA

```
output = Relu(BatchNormalization(input))
```

INHERITANCE

Inherits from [BatchNormalization](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::batch_norm
```

exNorm

Performs extended normalization computation, with optional gamma scaling and beta shift.

FORMULA

```
x = X
weight = W if W is not None else 1
bias = B if B is not None else 0

if mode == "layer":
    x_centered = x - np.mean(x, axis=axes, keepdims=True)
    y = x_centered * fast_inv_sqrt(
        np.mean(x_centered * x_centered, axis=axes, keepdims=True) + eps)

elif mode == "mean_var":
    mean = np.mean(x, axis=axes, keepdims=True)
    var = np.mean(x * x, axis=axes, keepdims=True) - mean * mean
    y = (x - mean) * fast_inv_sqrt(var + eps)

elif mode == "rms":
    y = x * fast_inv_sqrt(
        np.mean(x * x, axis=axes, keepdims=True) + eps)

elif mode == "pixel":
    y = x * fast_inv_sqrt(
        np.sum(x * x, axis=axes, keepdims=True) + eps)

elif mode == "l1":
    y = x / np.maximum(
        np.sum(np.abs(x), axis=axes, keepdims=True), eps)

elif mode == "l2":
    y = x / np.maximum(
        fast_sqrt(np.sum(x * x, axis=axes, keepdims=True)), eps)

elif mode == "rms_reciprocal":
    y = fast_inv_sqrt(
        np.mean(x * x, axis=axes, keepdims=True) + eps)

elif mode == "gaussian_topk":
    mean = np.mean(x, axis=axes, keepdims=True)
    std = fast_sqrt(np.mean(x * x, axis=axes, keepdims=True) - mean * mean)
    cutoff_x = mean + std * std_multipler
    y = np.clip(x - cutoff_x, a_min=0, a_max=None).astype(x.dtype)

Y = y * weight + bias
```

OPERATOR LEVEL

RKNN3 Extension Operator

PYTORCH SOURCE

```
aten::layer_norm, aten::rms_norm
```

INPUTS

- **X** - `T1` (required):
 - Shape: NCHW[batch, channel, height, width]

- Quantization: per-layer
- **W** - **T1** (optional):
 - Shape: NCHW[batch, channel, height, width], [1]
 - Quantization: per-layer
 - Gamma scale parameter; supports per-layer, per-channel, or broadcast shapes compatible with the output.
- **B** - **T1** (optional):
 - Shape: NCHW[batch, channel, height, width], [1]
 - Quantization: per-layer
 - Beta shift parameter; supports per-layer, per-channel, or broadcast shapes compatible with the output.

OUTPUTS

- **Y** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - When mode is rms_reciprocal, the dimensions specified by axes maintain keepdims semantics and output as 1.

ATTRIBUTES

- **mode - string**:
 - Valid values: layer, mean_var, rms, pixel, l1, l2, rms_reciprocal, gaussian_topk.
- **axes - int[axis0, axis1, axis2]**:
 - Default value is [1].
 - Shape inference supports negative axes; the NPU subgraph implementation primarily supports [1] and [1, 2, 3].
- **eps - float**:
 - Default value 1e-5, used to avoid division by zero.
- **std_multiplier - float**:
 - Only used by gaussian_topk; default value is 1.0.

DATA TYPE CONSTRAINTS

T1 in (**tensor<int8>** , **tensor<float16>** , **tensor<bfloat16>**):

exSoftmax

Applies softmax normalization along the specified axis of the input; the output shape is the same as the input.

FORMULA

```
exSoftmax(input, axis) = exp(input - max(input, axis)) / sum_axis(exp(input - max(input, axis)))
```

OPERATOR LEVEL

RKNN3 Extension Operator

PYTORCH SOURCE

```
aten::softmax
```

INPUTS

- **input** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Subject to the constraints of transpose specifications.

OUTPUTS

- **output** - **T1** :
 - Shape: NCHW[batch, channel, height, width]

ATTRIBUTES

- **axis** - **int**:
 - The axis for softmax; 1 is the channel direction, 3 is the width direction. Supporting only axis 1 and 3 is an RKNN platform limitation; the ONNX standard allows any valid axis.

DATA TYPE CONSTRAINTS

T1 in (**tensor<int8>**, **tensor<float16>**):

exSoftmaxMask

Masked Softmax computation: sets masked positions to -inf using mask_value, then applies Softmax.

FORMULA

```
input(x, mask_t, mask_value) = x + (mask_t - 1) * inf, if mask_value = 0  
x + mask_t * (-inf), if mask_value = 1
```

```
exSoftmaxMask(input, axis) = exp(input - max(input, axis)) / sum_axis(exp(input - max(input, axis)))
```

OPERATOR LEVEL

RKNN3 Extension Operator

PYTORCH SOURCE

```
aten::softmax
```

INPUTS

- **input** - `T1` (required):
 - Shape: NCHW[batch, channel, height, width]
 - Subject to the constraints of transpose specifications.
- **mask** - `T1` (optional):
 - Shape: NCHW[1, channel, height, width]
 - channel range: [1, 8192]
 - width range: [1, 8192]
 - When axis=1, width has no restriction; when axis=3/-1, width range is [1, 8192].

OUTPUTS

- **output** - `T1` :
 - Shape: NCHW[batch, channel, height, width]

ATTRIBUTES

- **axis** - `int`:
 - The axis for softmax; 1 is the channel direction, 3 is the width direction. Supporting only axis 1 and 3 is an RKNN platform limitation; the ONNX standard allows any valid axis.
- **mask_value** - `int`:
 - 0 indicates positions to keep, 1 indicates positions to mask.

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<float16>`):

Matrix Operations

exMatMul

Matrix multiplication operation.

FORMULA

$Y = A \cdot B + C$, where C is bias (bias is optional)

OPERATOR LEVEL

RKNN3 Extension Operator

PYTORCH SOURCE

`aten::matmul`, `aten::bmm`, `aten::mm`

INPUTS

- **A** - `T1` (required):
 - Shape: NK1N[batch, k, 1, n]
 - Quantization: per-layer, per-channel, group32/64/128
- **B** - `T1` (required):
 - Shape: NK1M[batch, k, 1, m]
 - Quantization: per-layer
- **C** - `T1`, `T2` (optional):
 - Shape: NM1M[batch, n, 1, m]
 - Quantization: per-layer
 - Supports NumPy-style broadcasting; broadcast constraints are the same as the Expand operator.

OUTPUTS

- **Y** - `T1`:
 - Shape: NM1M[batch, n, 1, m]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<float16>`): **T2** in (`const_tensor<float16>`):

exMatMulActivation

Fused exMatMul and generic Activation computation; activation type is determined by the Activation sub-operator.

FORMULA

```
output = Activation(exMatMul(input, weight, bias))
```

INHERITANCE

Inherits from [exMatMul](#).

OPERATOR LEVEL

RKNN3 Extension Operator

PYTORCH SOURCE

```
aten::matmul, aten::bmm, aten::mm
```

LLM Extension Operators

exAttention

General attention subgraph operator supporting FullAttention and SlidingAttention modes.

FORMULA

```
output = Matmul(Softmax(Matmul(rope(query, cos_pos_emb, sin_pos_emb), rope(key,
cos_pos_emb, sin_pos_emb)) + attention_mask), value)
output = Matmul(Softmax(Matmul(query, key) + attention_mask), value)
```

OPERATOR LEVEL

RKNN3 Extension Operator

PYTORCH SOURCE

```
aten::multi_head_attention_forward
```

INPUTS

- **query** - T1 (required):
 - Shape: NCHW[num_attention_heads, head_dim, 1, seq_len]
- **key** - T1 (required):
 - Shape: NCHW[num_key_value_heads, head_dim, 1, seq_len]
- **value** - T1 (required):
 - Shape: NCHW[num_key_value_heads, head_dim, 1, seq_len]
- **attention_mask** - T1 (required):
 - Shape: NCHW[1, seq_len, 1, seq_len]
- **cos_pos_emb** - T1 (optional):

- Shape: NCHW[1, head_dim, 1, max_position_embeddings]
- **sin_pos_emb** - T1 (optional):
 - Shape: NCHW[1, head_dim, 1, max_position_embeddings]
- **Th** - T2 (required):
 - Shape: N[1]
 - history token len; the number of tokens previously processed.
- **Tc** - T2 (required):
 - Shape: N[1]
 - current token len; the number of valid tokens in the current inference bucket.
- **Ts** - T2 (required):
 - Shape: N[1]
 - system prompt token len; the aligned system prompt length. Both [182X, 3572] require alignment to a multiple of 32.
- **Tsr** - T2 (required):
 - Shape: N[1]
 - Real system prompt token len; the actual system prompt length, less than or equal to Ts.

OUTPUTS

- **output** - T1:
 - Shape: NCHW[num_attention_heads, head_dim, 1, seq_len]

ATTRIBUTES

- **attention_type** - string:
 - Represents global attention and sliding window attention modes respectively. Valid values: [FullAttention, SlidingAttention], default FullAttention.
- **rope** - int:
 - Whether to use RoPE positional encoding; 0 means not used, 1 means used, default 0.
- **kvcache_store_method** - string:
 - Indicates the KV cache storage method; Normal stores float16 data directly, GroupQuant stores with group quantization. Valid values: [Normal, GroupQuant], default Normal.
- **kvcache_dtype** - string:
 - Indicates the KV cache data type. Valid values: [Float16, Int4_to_F8, Int4_to_F16]. Float16 uses float16 storage; Int4_to_F8 uses 4-bit integer quantization storage and converts to 8-bit float during computation; Int4_to_F16 uses 4-bit integer quantization storage and converts to 16-bit float during computation. Default Float16.

- **kvcache_buffer_len - int:**
 - Maximum length of KV cache, in tokens; must be less than the model's max_position_embeddings.
- **kvcache_group_size - int:**
 - Effective when kvcache_store_method is GroupQuant, indicating the group size for group quantization; when kvcache_store_method is Normal, defaults to 32.
- **kvcache_residual_depth - int:**
 - Effective when kvcache_store_method is GroupQuant, indicating that group quantization is triggered after storing this number of tokens; when kvcache_store_method is Normal, defaults to 32.
- **sliding_window_size - int:**
 - Effective when attention_type is SlidingAttention, indicating the sliding window size; when attention_type is FullAttention, defaults to -1.

DATA TYPE CONSTRAINTS

T1 in (tensor<float16>): **T2** in (tensor<int32>):

exSDPAttention

Performs scaled dot-product attention with optional mask.

FORMULA

```
q = Query
k = Key
v = Value

score = MatMul(k, q) * scale
if Mask is not None:
    score = score + Mask

prob = Softmax(score, axis=1)
Y = MatMul(v, prob)
```

OPERATOR LEVEL

RKNN3 Extension Operator

PYTORCH SOURCE

```
aten::scaled_dot_product_attention
```

INPUTS

- **Query** - **T1** (required):

- Shape: NCHW[batch, query_length, head, query_channel]
- Quantization: per-layer
- q matrix input, corresponding to query in the code.
- **Key** - `T1` (required):
 - Shape: NCHW[batch, key_length, head, key_channel]
 - Quantization: per-layer
 - k matrix input, corresponding to key in the code; participates in qk_matmul.
- **Value** - `T1` (required):
 - Shape: NCHW[batch, value_length, head, value_channel]
 - Quantization: per-layer
 - v matrix input.
- **Mask** - `T1` (optional):
 - Shape: NCHW[batch, key_channel, head, query_channel]
 - Quantization: per-layer
 - attention mask; when present, it is added to the qk_matmul result before softmax. The shape must be broadcastable to the qk_matmul output.

OUTPUTS

- **SDPAOutput** - `T1` :
 - Shape: NCHW[batch, query_length, head, value_channel]
 - Quantization: per-layer
 - Final output of scaled dot-product attention.

ATTRIBUTES

- **scale - float:**
 - Default value 1.0, applied to the qk_matmul result.
- **QMatmul - string:**
 - Quantization parameter description for qk_matmul output.
- **QAdd - string:**
 - Quantization parameter description for mask add output; only used when Mask is present.
- **QSoftmax - string:**
 - Quantization parameter description for softmax output.

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>` , `tensor<float16>` , `tensor<bfloat16>`):

exLinearAttention

Chunk Gated Delta Rule linear attention subgraph operator; computes recursive linear attention of query/key/value in chunks.

FORMULA

```
output = LinearAttention(query, key, value, g, beta, initial_state)
```

OPERATOR LEVEL

RKNN3 Extension Operator

INPUTS

- **query** - **T1** (required):
 - Shape: HD1S[H, D, 1, S]
 - Quantization: per-layer
 - H is num_heads, D is head_dim, S is seq_len.
- **key** - **T1** (required):
 - Shape: HD1S[H, D, 1, S]
 - Quantization: per-layer
- **value** - **T1** (required):
 - Shape: HD1S[H, D, 1, S]
 - Quantization: per-layer
- **g** - **T1** (required):
 - Shape: 1H1S[1, H, 1, S]
 - Quantization: per-layer
 - Gate factor.
- **beta** - **T1** (required):
 - Shape: 1H1S[1, H, 1, S]
 - Quantization: per-layer
- **Tc** - **T2** (required):
 - Shape: S[seq_len]
 - Effective seq_len scalar. When $Tc < seq_len$, left-side zero padding is applied to query/key/value along the seq dimension.
- **initial_state** - **T1** (optional):
 - Shape: HDD[H, D, 1, D]
 - Quantization: per-layer
 - Optional initial recurrent state. If provided by the frontend, it is directly connected.

OUTPUTS

- **output** - `T1`:
 - Shape: `HD1S[H, D, 1, S]`
 - Quantization: per-layer
- **final_state** - `T1`:
 - Shape: `HDD[H, D, 1, D]`
 - Quantization: per-layer
 - Only output when `OutputFinalState` is true.

ATTRIBUTES

- **ChunkSize** - `int`:
 - Chunk size, default 32. `seq_len` must be a multiple of `ChunkSize` (padding is the caller's responsibility).
- **UseQkL2Norm** - `bool`:
 - Whether to apply L2 normalization to query/key; default false.
- **OutputFinalState** - `bool`:
 - Whether to output the final recurrent state; default false.

DATA TYPE CONSTRAINTS

T1 in (`tensor<float16>` , `tensor<bfloat16>`): **T2** in (`tensor<int32>`):

NOTES

Subgraph operator; the actual implementation is expanded by `createSubGraph` at the IR layer; `emit` only returns success.

The batch dimension is fixed to 1. `seq_len` must be padded to a multiple of `ChunkSize` before entering the subgraph.

exRoPE

General subgraph operator supporting RoPE positional encoding.

FORMULA

```
output = input * cos_pos_emb + rotate_half(input) * sin_pos_emb
```

OPERATOR LEVEL

RKNN3 Extension Operator

INPUTS

- **input** - **T1** (required):
 - Shape: NCHW[batch_size, channel, 1, seq_len]
- **cos_pos_emb** - **T1** (required):
 - Shape: NCHW[batch_size, channel, 1, position_embeddings]
 - position_embeddings == 1 or seq_len <= position_embeddings <= max_position_embeddings。
- **sin_pos_emb** - **T1** (required):
 - Shape: NCHW[batch_size, channel, 1, position_embeddings]
 - position_embeddings == 1 or seq_len <= position_embeddings <= max_position_embeddings。
- **Th** - **T2** (optional):
 - Shape: N[1]
 - history token len; the number of tokens previously processed.
- **Tc** - **T2** (optional):
 - Shape: N[1]
 - current token len; the number of valid tokens in the current inference bucket.
- **Ts** - **T2** (optional):
 - Shape: N[1]
 - system prompt token len; the aligned system prompt length. Both [182X, 3572] require alignment to a multiple of 32.
- **Tsr** - **T2** (optional):
 - Shape: N[1]
 - Real system prompt token len; the actual system prompt length, less than or equal to Ts.

OUTPUTS

- **output** - **T1** :
 - Shape: NCHW[batch_size, channel, 1, seq_len]

DATA TYPE CONSTRAINTS

T1 in (**tensor<float16>**): **T2** in (**tensor<int32>**):

NOTES

When exRoPE is used in an LLM model and full cos/sin_pos_emb is provided (position_embeddings > seq_len), inputs such as Th/Tc/Ts/Tsr will be present. exRoPE internally calculates the actual length of cos/sin_pos_emb to use based on

Th/Tc/Ts/Tsr, thereby supporting dynamic sequence length inference. In other cases, seq_len == position_embeddings and Th/Tc/Ts/Tsr do not exist.

exSum

Weighted sum of multiple inputs, commonly used for merging LoRA multi-branch results.

FORMULA

```
Y = Sum(X_i * scale_i) for i in 0..N-1
```

OPERATOR LEVEL

RKNN3 Extension Operator

PYTORCH SOURCE

```
aten::sum
```

INPUTS

- **X** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - Variable-length input list; at least 1, all input shapes must be identical.

OUTPUTS

- **Y** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

ATTRIBUTES

- **scales** - **float[scale_1, ..., scale_N-1]**:
 - Weighting coefficient for each extra input (except the first); the count is the number of inputs - 1.

DATA TYPE CONSTRAINTS

T1 in (**tensor<float16>**):

I/O Processing

exInputProc

Performs input data preprocessing.

OPERATOR LEVEL

RKNN3 Extension Operator

INPUTS

- **A** - **T1** :
 - Shape: NCHW[batch, channel, height, width]

OUTPUTS

- **B** - **T2** :
 - Shape: NCHW[batch, channel, height, width]

DATA TYPE CONSTRAINTS

T1 in (tensor<float32> , tensor<float16> , tensor<bool> , tensor<uint8>): **T2** in (tensor<float16> , tensor<int8>):

exOutputProc

Performs output data post-processing.

OPERATOR LEVEL

RKNN3 Extension Operator

INPUTS

- **A** - **T1** :
 - Shape: NCHW[batch, channel, height, width]

OUTPUTS

- **B** - **T2** :
 - Shape: NCHW[batch, channel, height, width]

DATA TYPE CONSTRAINTS

T1 in (tensor<float16> , tensor<int8>): **T2** in (tensor<float32> , tensor<float16> , tensor<bool> , tensor<uint8>):

RK3572

Convolution

Conv

Convolution computation.

FORMULA

```
output = Conv(input_tensor, weight, bias)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::_convolution
```

INPUTS

- **input_tensor** - `T1` (required):
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
- **weight** - `T1` (required):
 - Shape: OIHW[output_channel, input_channel, kernel_h, kernel_w]
 - Quantization: per-layer, per-channel
 - kernel_h range: [1, 31]
 - kernel_w range: [1, 31]
- **bias** - `T1` (optional):
 - Shape: C[output_channel]
 - Quantization: per-layer

OUTPUTS

- **output** - `T1`:
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

ATTRIBUTES

- **strides** - `int[stride_h, stride_w]`:

- stride_h range: [1, 7]
- stride_w range: [1, 7]
- **pads** - `int[pads_top, pads_left, pads_bottom, pads_right]`:
 - pads_top range: [0, 15]
 - pads_left range: [0, 15]
 - pads_bottom range: [0, 15]
 - pads_right range: [0, 15]
- **group** - `int`: —
- **dilations** - `int[dilation_h, dilation_w]`:
 - dilation_h range: [1, 31]
 - dilation_w range: [1, 31]

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<float16>`):

SPECIAL CASES

Depthwise Convolution When Conv is used with depthwise convolution group semantics, the following differential specifications apply.

- **Input Differences** — **weight**: kernel_h range: [1, 8]; kernel_w range: [1, 8]
- **Attribute Differences** — **dilations**: dilation_h range: [1, 32]; dilation_w range: [1, 32]

ConvTranspose

Transposed convolution computation.

FORMULA

```
output = ConvTranspose(input_tensor, weight, bias)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::conv_transpose1d, aten::conv_transpose2d
```

INPUTS

- **input_tensor** - `T1` (required):
 - Shape: NCHW[batch, channel, height, width]

- Quantization: per-layer
- When `dilated_kernel_h > 1`, `input_tensor.width < 16383`.
The first layer input width must also satisfy the model input limitations.
- **weight** - `T1` (required):
 - Shape: OIHW[output_channel, input_channel, kernel_h, kernel_w]
 - Quantization: per-layer, per-channel
 - kernel_h range: [1, 31]
 - kernel_w range: [1, 31]
- **bias** - `T1` (optional):
 - Shape: C[output_channel]
 - Quantization: per-layer

OUTPUTS

- **output** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

ATTRIBUTES

- **strides** - `int[stride_h, stride_w]`:
 - stride_h range: [1, 8]
 - stride_w range: [1, 8]
- **pads** - `int[pads_top, pads_left, pads_bottom, pads_right]`:
 - pads_top range: [0, 15]
 - pads_left range: [0, 15]
 - pads_bottom range: [0, 15]
 - pads_right range: [0, 15]
 - `kernel_h * dilation_h - dilation_h - pads_top < 0` is not supported
 - `kernel_w * dilation_w - dilation_w - pads_left < 0` is not supported
 - `stride_h * (height - 1) - pads_top + 1 < output_h` is not supported
 - `stride_w * (width - 1) - pads_left + 1 < output_w` is not supported
- **group** - `int`:
 - Supports group = 1; when num_input = num_output, also supports group = num_output.
- **dilations** - `int[dilation_h, dilation_w]`:
 - dilation_h range: [1, 32]
 - dilation_w range: [1, 32]
 - When dilation != (1, 1), only kernel = (1, 1) is supported.

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<float16>`):

ConvActivation

Fused Conv and generic Activation computation; activation type is determined by the Activation sub-operator.

FORMULA

```
output = Activation(Conv(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [Conv](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::_convolution
```

ConvRelu

Fused Conv and Relu computation.

FORMULA

```
output = Relu(Conv(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [Conv](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::_convolution
```

ConvClip

Fused Conv and Clip computation.

FORMULA

```
output = Clip(Conv(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [Conv](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::_convolution
```

ConvSigmoid

Fused Conv and Sigmoid computation.

FORMULA

```
output = Sigmoid(Conv(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [Conv](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::_convolution
```

ConvTanh

Fused Conv and Tanh computation.

FORMULA

```
output = Tanh(Conv(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [Conv](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::_convolution
```

ConvElu

Fused Conv and Elu computation.

FORMULA

```
output = Elu(Conv(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [Conv](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::_convolution
```

ConvHardSigmoid

Fused Conv and HardSigmoid computation.

FORMULA

```
output = HardSigmoid(Conv(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [Conv](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::_convolution
```

ConvHardSwish

Fused Conv and HardSwish computation.

FORMULA

```
output = HardSwish(Conv(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [Conv](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::_convolution
```

ConvSwish

Fused Conv and Swish computation.

FORMULA

```
output = Swish(Conv(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [Conv](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::_convolution
```

ConvSoftplus

Fused Conv and Softplus computation.

FORMULA

```
output = Softplus(Conv(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [Conv](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::_convolution
```

ConvMish

Fused Conv and Mish computation.

FORMULA

```
output = Mish(Conv(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [Conv](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::_convolution
```

ConvAdd

Fused Conv and Add computation.

FORMULA

```
output = Add(Conv(input_tensor, weight, bias), residual)
```

INHERITANCE

Inherits from [Conv](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::_convolution
```

EXTRA INPUTS

- **residual** - `T1`:
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

- Second input of Add.
Supports broadcasting; broadcast constraints are the same as Add.

ConvAddRelu

Fused Conv, Add, and Relu computation.

FORMULA

```
output = Relu(Add(Conv(input_tensor, weight, bias), residual))
```

INHERITANCE

Inherits from [Conv](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::_convolution
```

EXTRA INPUTS

- **residual** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - Second input of Add.
Supports broadcasting; broadcast constraints are the same as Add.

ConvPRelu

Fused Conv and PRelu computation.

FORMULA

```
output = PRelu(Conv(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [Conv](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::_convolution
```

ConvTransposeActivation

Fused ConvTranspose and generic Activation computation; activation type is determined by the Activation sub-operator.

FORMULA

```
output = Activation(ConvTranspose(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [ConvTranspose](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::conv_transpose2d
```

ConvTransposeRelu

Fused ConvTranspose and Relu computation.

FORMULA

```
output = Relu(ConvTranspose(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [ConvTranspose](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::conv_transpose2d
```

ConvTransposeClip

Fused ConvTranspose and Clip computation.

FORMULA

```
output = Clip(ConvTranspose(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [ConvTranspose](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::conv_transpose2d
```

ConvTransposeSigmoid

Fused ConvTranspose and Sigmoid computation.

FORMULA

```
output = Sigmoid(ConvTranspose(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [ConvTranspose](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::conv_transpose2d
```

ConvTransposeTanh

Fused ConvTranspose and Tanh computation.

FORMULA

```
output = Tanh(ConvTranspose(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [ConvTranspose](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::conv_transpose2d
```

ConvTransposeElu

Fused ConvTranspose and Elu computation.

FORMULA

```
output = Elu(ConvTranspose(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [ConvTranspose](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::conv_transpose2d
```

ConvTransposeHardSigmoid

Fused ConvTranspose and HardSigmoid computation.

FORMULA

```
output = HardSigmoid(ConvTranspose(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [ConvTranspose](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::conv_transpose2d
```

ConvTransposeHardSwish

Fused ConvTranspose and HardSwish computation.

FORMULA

```
output = HardSwish(ConvTranspose(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [ConvTranspose](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::conv_transpose2d
```

ConvTransposeSwish

Fused ConvTranspose and Swish computation.

FORMULA

```
output = Swish(ConvTranspose(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [ConvTranspose](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::conv_transpose2d
```

ConvTransposeSoftplus

Fused ConvTranspose and Softplus computation.

FORMULA

```
output = Softplus(ConvTranspose(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [ConvTranspose](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::conv_transpose2d
```

ConvTransposeMish

Fused ConvTranspose and Mish computation.

FORMULA

```
output = Mish(ConvTranspose(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [ConvTranspose](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::conv_transpose2d
```

ConvTransposeAdd

Fused ConvTranspose and Add computation.

FORMULA

```
output = Add(ConvTranspose(input_tensor, weight, bias), residual)
```

INHERITANCE

Inherits from [ConvTranspose](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::conv_transpose2d
```

EXTRA INPUTS

- **residual** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

- Second input of Add.
Supports broadcasting; broadcast constraints are the same as Add.

ConvTransposePRelu

Fused ConvTranspose and PRelu computation.

FORMULA

```
output = PRelu(ConvTranspose(input_tensor, weight, bias))
```

INHERITANCE

Inherits from [ConvTranspose](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::conv_transpose2d
```

exConvStreaming

Convolution computation with streaming cache.

FORMULA

```
y = Conv(Concat(cache, x), w, b)
cache = Slice(Concat(cache, Slice(x, width=[0, tc])), width=[end - cache_w,
end])
```

OPERATOR LEVEL

RKNN3 Extension Operator

PYTORCH SOURCE

```
aten::_convolution
```

INPUTS

- **x** - `T1` (required):
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
- **w** - `T1` (required):

- Shape: OIHW[output_channel, input_channel, kernel_h, kernel_w]
- Quantization: per-layer, per-channel
- kernel_h range: [1, 31]
- kernel_w range: [1, 31]
- **b** - `T1` (optional):
 - Shape: C[output_channel]
 - Quantization: per-layer
 - Optional; when there is no bias but tc needs to be passed, b can be an empty input placeholder.
- **tc** - `T2` (optional):
 - Shape: N[1]
 - tc indicates the effective width written to cache this time; the length must be less than the input width.
When there are 3 inputs, if the size of the third input equals the output_channel of weight, it is treated as b; otherwise as tc.

OUTPUTS

- **y** - `T1`:
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

ATTRIBUTES

- **strides** - `int[stride_h, stride_w]`:
 - stride_h range: [1, 7]
 - stride_w range: [1, 7]
- **pads** - `int[pads_top, cache_w, pads_bottom, pads_right]`:
 - pads_top range: [0, 15]
 - pads_bottom range: [0, 15]
 - pads_right range: [0, 15]
 - cache_w is the streaming cache length; must be greater than 0.
- **group** - `int`: —
- **dilations** - `int[dilation_h, dilation_w]`:
 - dilation_h range: [1, 31]
 - dilation_w range: [1, 31]

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<float16>`): **T2** in (`tensor<int32>`):

SPECIAL CASES

Depthwise exConvStreaming When exConvStreaming is used with depthwise convolution group semantics, the following differential specifications apply.

- **Input Differences** — **w**: kernel_h range: [1, 8]; kernel_w range: [1, 8]
- **Attribute Differences** — **dilations**: dilation_h range: [1, 32]; dilation_w range: [1, 32]

NOTES

Subgraph operator; the actual implementation is expanded by createSubGraph at the IR layer; emit only returns success.

Pooling

AveragePool

Performs average pooling on the input tensor with the specified kernel_shape, strides, and pads.

FORMULA

```
Y = average_pool(X, kernel_shape, strides, pads)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::adaptive_avg_pool2d, aten::avg_pool2d
```

INPUTS

- **X** - T1:
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - int8/int16 quantization requires height <= 32 and width <= 32

OUTPUTS

- **Y** - T1:
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

ATTRIBUTES

- **kernel_shape** - `int[kernel_h, kernel_w]`:
 - kernel_h range: [1, 32]
 - kernel_w range: [1, 32]
- **strides** - `int[stride_h, stride_w]`:
 - stride_h range: [1, 63]
 - stride_w range: [1, 63]
- **pads** - `int[pad_top, pad_left, pad_bottom, pad_right]`:
 - pad_top range: [0, 63]
 - pad_left range: [0, 63]
 - pad_bottom range: [0, 63]
 - pad_right range: [0, 63]
- **auto_pad** - `string`:
 - Only explicit pads are supported.
- **ceil_mode** - `int`:
 - ceil 模式通过补充 bottom/right padding 支持。
- **count_include_pad** - `int`:
 - When set to 1, the mean calculation includes the padding area.
- **dilations** - `int[dilation_h, dilation_w]`:
 - dilation_h range: [1, 1]
 - dilation_w range: [1, 1]

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<float16>`, `tensor<bfloat16>`):

MaxPool

Performs max pooling on the input tensor with the specified kernel_shape, strides, and pads.

FORMULA

```
Y = max_pool(X, kernel_shape, strides, pads)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

`aten::max_pool1d`, `aten::max_pool2d`

INPUTS

- **X - T1**:
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **Y - T1**:
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

ATTRIBUTES

- **kernel_shape - int[kernel_h, kernel_w]**:
 - kernel_h range: [1, 32]
 - kernel_w range: [1, 256]
 - kernel_h * kernel_w <= 256
- **strides - int[stride_h, stride_w]**:
 - stride_h range: [1, 31]
 - stride_w range: [1, 256]
- **pads - int[pad_top, pad_left, pad_bottom, pad_right]**:
 - pad_top range: [0, 63]
 - pad_left range: [0, 63]
 - pad_bottom range: [0, 63]
 - pad_right range: [0, 63]
- **auto_pad - string**:
 - Only explicit pads are supported.
- **ceil_mode - int**:
 - ceil 模式通过补充 bottom/right padding 支持。
- **dilations - int[dilation_h, dilation_w]**:
 - dilation_h range: [1, 1]
 - dilation_w range: [1, 1]

DATA TYPE CONSTRAINTS

T1 in (tensor<int8> , tensor<float16>): **T2** in (tensor<int64>):

GlobalAveragePool

Performs global average pooling on the H/W spatial dimensions of the input tensor.

FORMULA

```
Y = mean(X, axes=[2, 3])
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::adaptive_avg_pool2d
```

INPUTS

- **X** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - int8 quantization requires height <= 32 and width <= 32;

OUTPUTS

- **Y** - **T1** :
 - Shape: NCHW[batch, channel, 1, 1]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (tensor<int8> , tensor<float16> , tensor<bfloat16>):

GlobalMaxPool

Performs global max pooling on the spatial dimensions of the input tensor, outputting the maximum value per channel.

FORMULA

```
Y[batch, channel, 0, 0] = max(X[batch, channel, :, :])
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::adaptive_max_pool2d
```

INPUTS

- **X** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - Hard constraint: height * width <= 256 and height <= 32.

OUTPUTS

- **Y** - **T1** :
 - Shape: NCHW[batch, channel, 1, 1]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<float16>`):

Elementwise Operations

Add

Performs element-wise binary addition, supporting NumPy-style broadcasting.

FORMULA

```
C = A + B
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::add , aten::add_
```

INPUTS

- **A** - **T1, T2** :
 - Shape: NCHW[batch, channel, height, width]

- Quantization: per-layer
- Supports NumPy-style broadcasting; broadcast constraints are the same as the Expand operator.
- **B** - `T1, T2` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - Supports NumPy-style broadcasting; broadcast constraints are the same as the Expand operator.

OUTPUTS

- **C** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>` , `tensor<float16>`): **T2** in (`const_tensor<float16>`):

Sub

Performs element-wise binary subtraction, supporting NumPy-style broadcasting.

FORMULA

`C = A - B`

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

`aten::sub` , `aten::sub_`

INPUTS

- **A** - `T1, T2` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - Supports NumPy-style broadcasting; broadcast constraints are the same as the Expand operator.
- **B** - `T1, T2` :
 - Shape: NCHW[batch, channel, height, width]

- Quantization: per-layer
- Supports NumPy-style broadcasting; broadcast constraints are the same as the Expand operator.

OUTPUTS

- **C** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (tensor<int8> , tensor<float16>): **T2** in (const_tensor<float16>):

Mul

Performs element-wise binary multiplication, supporting NumPy-style broadcasting.

FORMULA

$C = A * B$

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

aten::mul , aten::mul_

INPUTS

- **A** - **T1, T2** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - Supports NumPy-style broadcasting; broadcast constraints are the same as the Expand operator.
- **B** - **T1, T2** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - Supports NumPy-style broadcasting; broadcast constraints are the same as the Expand operator.

OUTPUTS

- **C** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>` , `tensor<float16>`): **T2** in (`const_tensor<float16>`):

Div

Performs element-wise binary division, supporting NumPy-style broadcasting.

FORMULA

$$C = A / B$$

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

`aten::div` , `aten::true_divide`

INPUTS

- **A** - **T1, T2** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - Supports NumPy-style broadcasting; broadcast constraints are the same as the Expand operator.
- **B** - **T1, T2** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - Supports NumPy-style broadcasting; broadcast constraints are the same as the Expand operator.

OUTPUTS

- **C** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (tensor<float16>): **T2** in (const_tensor<float16>):

Min

Computes the element-wise minimum of multiple input tensors, supporting NumPy-style broadcasting.

FORMULA

```
C = min(A, B, ...)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::min, aten::minimum
```

INPUTS

- **A** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - Supports NumPy-style broadcasting; broadcast constraints are the same as the Expand operator.
- **B** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - Supports NumPy-style broadcasting; broadcast constraints are the same as the Expand operator.

OUTPUTS

- **C** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer, per-channel

DATA TYPE CONSTRAINTS

T1 in (tensor<int8>, tensor<float16>):

Max

Computes the element-wise maximum of multiple input tensors, supporting NumPy-style broadcasting.

FORMULA

```
C = max(A, B, ...)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::max, aten::maximum
```

INPUTS

- **A** - **T1**:
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - Supports NumPy-style broadcasting; broadcast constraints are the same as the Expand operator.
- **B** - **T1**:
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - Supports NumPy-style broadcasting; broadcast constraints are the same as the Expand operator.

OUTPUTS

- **C** - **T1**:
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer, per-channel

DATA TYPE CONSTRAINTS

T1 in (tensor<int8>, tensor<float16>):

Pow

Performs element-wise exponentiation.

FORMULA

```
output = input ^ exponent
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::pow , aten::pow_
```

INPUTS

- **input** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
- **exponent** - `T2` :
 - Shape: S[scalar]

OUTPUTS

- **output** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>` , `tensor<float16>`): **T2** in (`const_tensor<float16>`):

Neg

Performs element-wise negation on the input tensor.

FORMULA

```
Y = -X
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::neg , aten::neg_
```

INPUTS

- **X** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **Y** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>` , `tensor<float16>`):

Exp

Computes the element-wise base-e exponential of the input tensor.

FORMULA

```
output = exp(input)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::exp , aten::exp_
```

INPUTS

- **input** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **output** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (tensor<int8> , tensor<float16>):

Sqrt

Performs element-wise square root on the input tensor.

FORMULA

```
output = sqrt(input)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::sqrt , aten::sqrt_
```

INPUTS

- **input** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **output** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (tensor<int8> , tensor<float16>):

Reciprocal

Performs element-wise reciprocal on the input tensor.

FORMULA

```
Y = 1 / X
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::reciprocal, aten::reciprocal_
```

INPUTS

- **X** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **Y** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (tensor<int8>, tensor<float16>):

Floor

Performs element-wise floor (rounds towards negative infinity) on the input tensor.

FORMULA

```
output = floor(input)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::floor, aten::floor_
```

INPUTS

- **input** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **output** - **T1** :
 - Shape: NCHW[batch, channel, height, width]

- Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<float16>`):

Round

Rounding function using round-half-up semantics, formula: $y = \text{Floor}(x + 0.5)$. Differs from ONNX Round's round-half-to-even (banker's rounding).

FORMULA

`y = Floor(x + 0.5)`

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

`aten::round`, `aten::round_`

INPUTS

- **x** - **T1**:
 - Shape: NCHW[batch, channel, height, width]

OUTPUTS

- **y** - **T1**:
 - Shape: NCHW[batch, channel, height, width]

DATA TYPE CONSTRAINTS

T1 in (`tensor<float16>`):

Sin

Computes the element-wise sine function.

FORMULA

`y = sin(x)`

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

`aten::sin` , `aten::sin_`

INPUTS

- **input** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **output** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>` , `tensor<float16>`):

NOTES

Implementation first applies $\text{mod } 2\pi$ to the input, then performs a LUT lookup.

Cos

Computes the element-wise cosine function.

FORMULA

$y = \cos(x)$

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

`aten::cos` , `aten::cos_`

INPUTS

- **input** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **output** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<float16>`):

NOTES

Implementation first applies $\text{mod } 2\pi$ to the input, then performs a LUT lookup.

exFastExp

Computes the element-wise natural exponential using an approximation for improved performance.

FORMULA

`output  $\approx$  exp(input)`

OPERATOR LEVEL

RKNN3 Extension Operator

PYTORCH SOURCE

`aten::exp`, `aten::exp_`

INPUTS

- **input** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **output** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<float16>`):

exTwoPow

Computes the element-wise power of 2.

FORMULA

$$y = 2^x$$

OPERATOR LEVEL

RKNN3 Extension Operator

PYTORCH SOURCE

```
aten::pow
```

INPUTS

- **input** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **output** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<float16>`):

AddRelu

Fused Add and Relu computation.

FORMULA

$$C = \text{Relu}(A + B)$$

INHERITANCE

Inherits from [Add](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::add , aten::add_
```

MulRelu

Fused Mul and Relu computation.

FORMULA

```
C = Relu(A * B)
```

INHERITANCE

Inherits from [Mul](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::mul , aten::mul_
```

Activation Functions

Relu

Rectified Linear Unit, performs $\max(\text{input}, 0)$ element-wise.

FORMULA

```
output = max(input, 0)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::relu , aten::relu6 , aten::relu_
```

INPUTS

- **input** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **output** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>` , `tensor<float16>`):

Sigmoid

Applies the Sigmoid function element-wise to the given input tensor.

FORMULA

$$y = 1 / (1 + \exp(-x))$$

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

`aten::sigmoid` , `aten::sigmoid_`

INPUTS

- **input** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **output** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>` , `tensor<float16>`):

Clip

Clips the input tensor elements to the [min, max] range; out-of-bounds values are clamped to the boundary.

FORMULA

```
output = Clip(input, min, max)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::clamp, aten::clamp_
```

INPUTS

- **input** - `T1` (required):
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
- **min** - `T2` (optional):
 - Shape: S[scalar]
 - Clip lower bound, scalar. Omitted means no lower bound.
- **max** - `T2` (optional):
 - Shape: S[scalar]
 - Clip upper bound, scalar. Omitted means no upper bound.

OUTPUTS

- **output** - `T1`:
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<float16>`): **T2** in (`const_tensor<float16>`):

LeakyRelu

Leaky Rectified Linear Unit; positive values remain unchanged, negative values are scaled by a fixed ratio alpha.

FORMULA

```
output = input if input >= 0 else alpha * input
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::leaky_relu, aten::leaky_relu_
```

INPUTS

- **input** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **output** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

ATTRIBUTES

- **alpha** - `float[alpha]`:
 - alpha range: [0, 1]
 - Negative value scaling factor.

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<float16>`):

PRelu

Parametric Rectified Linear Unit; positive values remain unchanged, negative values are scaled by a learnable parameter slope.

FORMULA

```
output = input if input >= 0 else slope * input
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::prelu, aten::prelu_
```

INPUTS

- **input** - `T1` :

- Shape: NCHW[batch, channel, height, width]
- Quantization: per-layer, per-channel
- **slope** - `T2` :
 - Shape: C[channel]
 - Quantization: per-channel
 - Supports a single scalar or C-dimension coefficients.

OUTPUTS

- **output** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer, per-channel

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<float16>`): **T2** in (`const_tensor<float16>`):

Elu

Applies the Elu function element-wise, alpha = 1.

FORMULA

$f(x) = \alpha * (\exp(x) - 1) \text{ for } x < 0; f(x) = x \text{ for } x \geq 0$

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

`aten::elu`, `aten::elu_`

INPUTS

- **input** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **output** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (tensor<int8> , tensor<float16>):

Softplus

Applies the Softplus function element-wise.

FORMULA

$$y = \ln(\exp(x) + 1)$$

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

aten::softplus

INPUTS

- **X** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **Y** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (tensor<int8> , tensor<float16>):

HardSigmoid

Applies the HardSigmoid function element-wise, controlled by alpha and beta attributes.

FORMULA

$$y = \max(0, \min(1, \alpha * x + \beta))$$

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::hardsigmoid, aten::hardsigmoid_
```

INPUTS

- **input** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **output** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

ATTRIBUTES

- **alpha** - **float**:
 - Default 0.2 (consistent with the ONNX standard).
- **beta** - **float**:
 - Default 0.5.

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>` , `tensor<float16>`):

HardSwish

Applies the HardSwish function element-wise, $\alpha = 1/6$, $\beta = 0.5$.

FORMULA

```
y = x * max(0, min(1, alpha * x + beta))
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::hardswish, aten::hardswish_
```

INPUTS

- **X** - `T1` :

- Shape: NCHW[batch, channel, height, width]
- Quantization: per-layer

OUTPUTS

- **Y** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (tensor<int8> , tensor<float16>):

Tanh

Computes the element-wise hyperbolic tangent of the given input tensor.

FORMULA

$y = \tanh(x)$

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

aten::tanh , aten::tanh_

INPUTS

- **X** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **Y** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (tensor<int8> , tensor<float16>):

Mish

Applies the Mish function element-wise.

FORMULA

$$y = x * \tanh(\text{softplus}(x)) = x * \tanh(\ln(1 + \exp(x)))$$

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::mish
```

INPUTS

- **input** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **output** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<float16>`):

exGelu

Applies the Gelu function element-wise.

FORMULA

$$y = x * \Phi(x), \text{ when } \Phi \text{ uses } \tanh \text{ approximation: } y = 0.5 * x * (1 + \tanh(\sqrt{2/\pi} * (x + 0.044715 * x^3)))$$

OPERATOR LEVEL

RKNN3 Extension Operator

PYTORCH SOURCE

```
aten::gelu
```

INPUTS

- **input** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **output** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>` , `tensor<float16>`):

exHardSwish

Applies the HardSwish function element-wise, $\alpha = 1/6$, $\beta = 0.5$.

FORMULA

```
y = x * max(0, min(1, alpha * x + beta))
```

OPERATOR LEVEL

RKNN3 Extension Operator

PYTORCH SOURCE

```
aten::hardswish , aten::hardswish_
```

INPUTS

- **X** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **Y** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (tensor<int8> , tensor<float16>):

exMish

Applies the Mish function element-wise.

FORMULA

$$y = x * \tanh(\text{softplus}(x)) = x * \tanh(\ln(1 + \exp(x)))$$

OPERATOR LEVEL

RKNN3 Extension Operator

PYTORCH SOURCE

```
aten::mish , aten::mish_
```

INPUTS

- **input** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **output** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (tensor<int8> , tensor<float16>):

exSwish

Applies the Swish function (also known as SiLU) element-wise.

FORMULA

$$y = x * \text{sigmoid}(x)$$

OPERATOR LEVEL

RKNN3 Extension Operator

PYTORCH SOURCE

```
aten::silu, aten::silu_
```

INPUTS

- **X** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **Y** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (tensor<int8>, tensor<float16>):

exGlu

Splits the input tensor into two equal parts along the specified axis, applies sigmoid gating to the second part, and multiplies it with the first part.

FORMULA

```
a, b = np.split(X, 2, axis=axis)
Y = a * sigmoid(b)
```

OPERATOR LEVEL

RKNN3 Extension Operator

PYTORCH SOURCE

```
aten::glu
```

INPUTS

- **X** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - The length of the dimension specified by axis must be even.
The NPU subgraph implementation primarily targets axis = 1 channel dimension

splitting.

OUTPUTS

- **Y - T1 :**
 - Shape: NCHW[batch, out_channel, height, width]
 - Quantization: per-layer
 - The output shape is consistent with the input shape, but the dimension specified by axis becomes half of the input.
When axis = 1, out_channel = channel / 2.

ATTRIBUTES

- **axis - int:**
 - Specifies the dimension for GLU split; valid range is [-rank, rank - 1].
Negative axes are converted to positive axes following NumPy semantics.

DATA TYPE CONSTRAINTS

T1 in (tensor<int8>, tensor<float16>):

exLogit

Applies the logit activation function to the input tensor.

FORMULA

$$Y = \log(X / (1 - X))$$

OPERATOR LEVEL

RKNN3 Extension Operator

PYTORCH SOURCE

```
aten::logit
```

INPUTS

- **X - T1 :**
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **Y - T1 :**

- Shape: NCHW[batch, channel, height, width]
- Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>` , `tensor<float16>`):

Comparison & Logical Operations

Equal

Compares whether two tensors are equal element-wise, outputs a bool result, supporting NumPy-style broadcasting.

FORMULA

`C = (A == B)`

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

`aten::eq` , `aten::eq_`

INPUTS

- **A** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
- **B** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **C** - `T2` :
 - Shape: NCHW[batch, channel, height, width]
 - Boolean semantic result; RKNN stores it in int8 memory layout.

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>` , `tensor<float16>`) : **T2** in (`tensor<bool>` , `tensor<int8>`):

Greater

Compares whether A is greater than B element-wise, outputs a bool result, supporting NumPy-style broadcasting.

FORMULA

$$C = (A > B)$$

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

`aten::gt`, `aten::gt_`

INPUTS

- **A** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - Supports NumPy-style broadcasting.
- **B** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - Supports NumPy-style broadcasting.

OUTPUTS

- **C** - `T2` :
 - Shape: NCHW[batch, channel, height, width]

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<float16>`): **T2** in (`tensor<bool>`, `tensor<int8>`):

GreaterOrEqual

Compares whether A is greater than or equal to B element-wise, outputs a bool result, supporting NumPy-style broadcasting.

FORMULA

$$C = (A \geq B)$$

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

`aten::ge` , `aten::ge_`

INPUTS

- **A** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
- **B** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **C** - `T2` :
 - Shape: NCHW[batch, channel, height, width]
 - Boolean semantic result; RKNN stores it in int8 memory layout.

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>` , `tensor<float16>`): **T2** in (`tensor<bool>` , `tensor<int8>`):

Less

Compares whether A is less than B element-wise, outputs a bool result, supporting NumPy-style broadcasting.

FORMULA

`C = (A < B)`

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

`aten::lt` , `aten::lt_`

INPUTS

- **A** - `T1` :

- Shape: NCHW[batch, channel, height, width]
- Quantization: per-layer
- Supports NumPy-style broadcasting.
- **B** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - Supports NumPy-style broadcasting.

OUTPUTS

- **C** - **T2** :
 - Shape: NCHW[batch, channel, height, width]

DATA TYPE CONSTRAINTS

T1 in (tensor<int8> , tensor<float16>): **T2** in (tensor<bool> , tensor<int8>):

LessOrEqual

Compares whether A is less than or equal to B element-wise, outputs a bool result, supporting NumPy-style broadcasting.

FORMULA

$C = (A \leq B)$

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

aten::le , aten::le_

INPUTS

- **A** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - Supports NumPy-style broadcasting.
- **B** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

- Supports NumPy-style broadcasting.

OUTPUTS

- **C** - **T2** :
 - Shape: NCHW[batch, channel, height, width]

DATA TYPE CONSTRAINTS

T1 in (tensor<int8> , tensor<float16>): **T2** in (tensor<bool> , tensor<int8>):

Not

Performs element-wise logical NOT on the input tensor.

FORMULA

$$Y = !X$$

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

aten::logical_not

INPUTS

- **X** - **T1** :
 - Shape: NCHW[batch, channel, height, width]

OUTPUTS

- **Y** - **T1** :
 - Shape: NCHW[batch, channel, height, width]

DATA TYPE CONSTRAINTS

T1 in (tensor<bool>):

And

Performs element-wise logical AND, outputs a bool result, supporting NumPy-style broadcasting.

FORMULA

```
C = A && B
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::bitwise_and, aten::logical_and
```

INPUTS

- **A** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Supports NumPy-style broadcasting; broadcast constraints are the same as the Expand operator.
- **B** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Supports NumPy-style broadcasting; broadcast constraints are the same as the Expand operator.

OUTPUTS

- **C** - `T1` :
 - Shape: NCHW[batch, channel, height, width]

DATA TYPE CONSTRAINTS

T1 in (`tensor<bool>`, `tensor<int8>`):

Where

Selects elements from x or y based on the condition, supporting NumPy-style broadcasting.

FORMULA

```
output = select(condition, x, y)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::where
```

INPUTS

- **condition** - T1 :
 - Shape: NCHW[batch, channel, height, width]
- **x** - T2 :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - Supports NumPy-style broadcasting.
- **y** - T2 :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - Supports NumPy-style broadcasting.

OUTPUTS

- **output** - T2 :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (tensor<bool>): **T2** in (tensor<int8> , tensor<float16>):

Reduction Operations

ReduceMax

Computes the maximum of the input tensor along the specified axis, supporting multiple axes and keepdims.

FORMULA

```
output = ReduceMax(input, axes, keepdims)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::amax , aten::max
```

INPUTS

- **input** - T1 :

- Shape: NCHW[batch, channel, height, width]
- Quantization: per-layer

OUTPUTS

- **output** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - When keepdims=1, the reduced axis has size 1; when keepdims=0, the axis is removed from the output. The actual output rank depends on axes and keepdims.

ATTRIBUTES

- **axes** - `int[]`:
 - Axis for computing the maximum value.
- **keepdims** - `int`:
 - Whether to keep the reduced dimensions (1 keeps, 0 removes).

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<float16>`):

ReduceMean

Reduces the input tensor by computing the mean along the specified axes.

FORMULA

```
Y = mean(data, axes)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::mean
```

INPUTS

- **data** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - For int8/int16 quantization with axes = [2], [3], [2, 3], or [1, 2, 3], height <= 32 and

width ≤ 32 are required

- **axes** - `T2` :
 - Shape: Axes[num_axes]
 - Specifies the dimensions to reduce; negative axes are converted to positive axes following NumPy semantics.
Supports axes = [1], [2], [3], [2, 3], [1, 2, 3]

OUTPUTS

- **reduced** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (tensor<int8>, tensor<float16>, tensor<bfloat16>): **T2** in (const_tensor<int64>):

ReduceSum

Reduces the input tensor by computing the sum along the specified axes.

FORMULA

`Y = sum(data, axes)`

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

`aten::sum`

INPUTS

- **data** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - For int8/int16 quantization with axes = [2], [3], [2, 3], or [1, 2, 3], height ≤ 32 and width ≤ 32 are required
- **axes** - `T2` :
 - Shape: Axes[num_axes]

- Specifies the dimensions to reduce; negative axes are converted to positive axes following NumPy semantics.
Supports axes = [1], [2], [3], [2, 3], [1, 2, 3]

OUTPUTS

- **reduced** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (tensor<int8> , tensor<float16> , tensor<bfloat16>): **T2** in (const_tensor<int64>):

ArgMax

Returns the indices of the maximum values along the specified axis.

FORMULA

```
output = ArgMax(input, axis)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::argmax
```

INPUTS

- **input** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **output** - **T2** :
 - Shape: NCHW[batch, channel, height, width]

ATTRIBUTES

- **axis** - **int**:
 - Specifies the axis for argmax operation; valid range [0, ndim-1].

- **keepdims - int:**
 - Whether to keep the output dimensions the same as the input. 0 means no, 1 means yes.
- **select_last_index - int:**
 - When multiple maximum values exist, whether to select the last index. 0 selects the first, 1 selects the last.

DATA TYPE CONSTRAINTS

T1 in (tensor<int8> , tensor<float16>): **T2** in (tensor<int32>):

ArgMin

Returns the indices of the minimum values along the specified axis.

FORMULA

```
output = ArgMin(input, axis)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::argmin
```

INPUTS

- **input - T1 :**
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **output - T2 :**
 - Shape: NCHW[batch, channel, height, width]

ATTRIBUTES

- **axis - int:**
 - Specifies the axis for argmin operation; valid range [0, ndim-1].
- **keepdims - int:**
 - Whether to keep the output dimensions the same as the input. 0 means no, 1 means yes.

- **select_last_index - int:**

- When multiple minimum values exist, whether to select the last index. 0 selects the first, 1 selects the last.

DATA TYPE CONSTRAINTS

T1 in (tensor<int8>, tensor<float16>): **T2** in (tensor<int32>):

TopK

Retrieves the K largest (or smallest, determined by largest) values and their indices along the specified axis.

FORMULA

```
(values, indices) = TopK(X, K, axis, largest, sorted)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::topk
```

INPUTS

- **X** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
- **K** - **T3** :
 - Shape: S[scalar]
 - The top K values, scalar, stored in an int64 tensor.

OUTPUTS

- **values** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - The top K values.
- **indices** - **T2** :
 - Shape: NCHW[batch, channel, height, width]
 - Index of the corresponding value. The ONNX standard is int64; RKNN actually outputs int32 or int64.

ATTRIBUTES

- **axis - int:**
 - Axis for computation; default -1.
- **largest - int:**
 - 1 selects maximum, 0 selects minimum, default 1.
- **sorted - int:**
 - 1 outputs in descending order, 0 does not guarantee order, default 1.

DATA TYPE CONSTRAINTS

T1 in (tensor<int8> , tensor<float16>): **T2** in (tensor<int32> , tensor<int64>): **T3** in (tensor<int64>):

Tensor Transform

Reshape

Returns a tensor with the specified shape without changing the input data or element count.

FORMULA

```
output = reshape(input, shape)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::reshape , aten::reshape_
```

INPUTS

- **input - T1 :**
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **output - T1 :**
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

ATTRIBUTES

- **shape - int:**
 - Specifies the output tensor shape.

DATA TYPE CONSTRAINTS

T1 in (tensor<int8> , tensor<float16>):

Transpose

Transposes the input tensor according to the axis order specified by perm.

FORMULA

```
Y = transpose(X, perm)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::permute , aten::transpose , aten::transpose_
```

INPUTS

- **X - T1, T2 :**
 - Shape: NCHW[n1, c1, h1, w1]
 - Quantization: per-layer
 - For detailed constraint rules, see the perm attribute list.

OUTPUTS

- **Y - T1, T2 :**
 - Shape: NCHW[n2, c2, h2, w2]
 - Quantization: per-layer
 - For detailed constraint rules, see the perm attribute list.

ATTRIBUTES

- **perm - int[d0, d1, d2, d3]:**
 - Axis order for transpose. Constraints for different perm parameters are listed in the perm constraint table below.

DATA TYPE CONSTRAINTS

T1 in (tensor<int8>): **T2** in (tensor<int16> , tensor<float16>):

NOTES

Different perm parameter constraints are as follows (where $p = 268435456$, $q = 65536$):

perm	T1 Constraint	T2 Constraint
[0,1,2,3]	No restriction	No restriction
[0,1,3,2]	$h1\%16=0, h1w1<16p, h1w1<2q$	$h1\%8=0, h1w1<8p, h1w1<2q$
[0,2,3,1]	$c1<8192, h1w1\%8=0, w1c1<p*16$	$c1<8192, h1w1\%8=0, w1c1<p*8$
[0,2,1,3]	$h1\%16=0, h1w1<16p, h1w1<2q, c1*w1<p$	$h1\%8=0, h1w1<8p, h1w1<2q, c1*w1<p$
[0,3,1,2]	$h1w1<p, w1<8192, h1c1<p$	$h1w1<p, w1<8192, h1c1<p$
[0,3,2,1]	$h1\%16=0, h1w1<16p, h1w1<2q, c1<p*16$	$h1\%8=0, h1w1<8p, h1w1<2q, c1<p*8$
[1,0,2,3]	$h1w1<16p, n1<8192, h1*w1<p$	$h1w1<8p, n1<8192, h1*w1<p$
[1,0,3,2]	$h1\%16=0, c1h1w1<16p, h1w1<2*q$	$h1\%8=0, c1h1w1<8p, h1w1<2*q$
[1,2,0,3]	$n1w1<p16, c1h1w1<p, w1\%16=0$	$n1w1<p8, c1h1w1<p, w1\%8=0$
[1,2,3,0]	$h1w1<p, n1<p16, c1h1w1<p16, n1w1<p*16$	$h1w1<p, n1<p8, c1h1w1<p8, n1w1<p*8$
[1,3,0,2]	$c1w1<8192, n1h1<p, w1\%16=0$	$c1w1<8192, n1h1<p, w1\%8=0$
[1,3,2,0]	$c1h1w1<p, n1*h1<p, w1<8192, w1\%16=0$	$c1h1w1<p, n1*h1<p, w1<8192, w1\%8=0$
[2,1,0,3]	$c1<8192, c1h1w1<p, n1*w1<p$	$c1<8192, c1h1w1<p, n1*w1<p$
[2,1,3,0]	$c1h1w1<p, n1w1<p16, c1<8192, n1*w1\%4=0$	$c1h1w1<p, n1w1<p8, c1<8192, n1*w1\%4=0$
[2,0,3,1]	$c1<8192, c1h1w1<p$	$c1<8192, c1h1w1<p$
[2,0,1,3]	$n1c1h1w1<p, h1w1\%16=0$	$n1c1h1w1<p, h1w1\%8=0$
[2,3,1,0]	$c1h1w1<p, n1c1<p, h1w1<8192, h1*w1\%16=0$	$c1h1w1<p, n1c1<p, h1w1<8192, h1*w1\%8=0$
[2,3,0,1]	$h1w1<8192, n1c1<p$	$h1w1<8192, n1c1<p$
[3,0,2,1]	$h1w1<p, n1c1<8192, c1h1<p, n1w1<8192, n1*w1\%16=0$	$h1w1<p, n1c1<8192, c1h1<p, n1w1<8192, n1*w1\%8=0$
[3,0,1,2]	$h1w1<p, n1c1<8192, c1h1<p, n1w1<8192, n1h1w1\%16=0$	$h1w1<p, n1c1<8192, c1h1<p, n1w1<8192, n1h1w1\%8=0$
[3,2,0,1]	$h1w1<p, n1c1<8192, n1c1w1<p$	$h1w1<p, n1c1<8192, n1c1w1<p$
[3,2,1,0]	$c1h1w1<p, h1w1<8192, 2h1w1<q, n1c1<p*16, h1\%16=0$	$c1h1w1<p, h1w1<8192, 2h1w1<q, n1c1<p*8, h1\%8=0$
[3,1,0,2]	$h1w1<p, c1w1<8192, n1*h1<p$	$h1w1<p, c1w1<8192, n1*h1<p$
[3,1,2,0]	$h1w1<p, n1c1<8192, n1h1<p, n1c1*w1<p$	$h1w1<p, n1c1<8192, n1h1<p, n1c1*w1<p$

Concat

Concatenates multiple tensors along the specified axis into a single tensor; all inputs must have the same size except along the axis dimension.

FORMULA

```
output = Concat(input_tensor_list, axis)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::cat
```

INPUTS

- **input_tensor** - **T**:
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - Input is a list of tensors; all tensors must have the same size except along the axis dimension.

OUTPUTS

- **output** - **T**:
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

ATTRIBUTES

- **axis** - **int[axis]**:
 - axis range: [-4, 3]
 - Concatenation axis; RKNN requires 4D NCHW input, supports negative axes.

DATA TYPE CONSTRAINTS

T in (**tensor<int8>**, **tensor<float16>**):

Split

Splits the input tensor into multiple output tensors along the specified axis.

FORMULA

```
outputs = Split(input_tensor, axis, num_outputs, split)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::split
```

INPUTS

- **input_tensor** - **T**:
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **output** - **T**:
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - One or more output tensors; the count is determined by num_outputs, and the shape depends on the split.

ATTRIBUTES

- **axis** - **int[axis]**:
 - axis range: [-4, 3]
 - Split axis; RKNN requires 4D NCHW input, supports negative axes.
- **num_outputs** - **int**: —
- **split** - **int[]**:
 - Specifies the size of each output along the axis dimension after splitting.

DATA TYPE CONSTRAINTS

T in (`tensor<int8>`, `tensor<float16>`):

Expand

Broadcasts the input tensor to the target shape.

FORMULA

```
output = Expand(input, target_shape)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::expand , aten::expand_as
```

INPUTS

- **input** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **output** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>` , `tensor<float16>`):

Slice

Obtains a slice of the input tensor following NumPy slicing rules.

FORMULA

```
output = Slice(data, starts, ends, axes, steps)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::narrow , aten::slice
```

INPUTS

- **data** - `T` (required):
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
- **starts** - `T2` (required):

- Shape: D[dim]
- Per-axis split start positions.
- **ends** - T2 (required):
 - Shape: D[dim]
 - Per-axis split end positions.
- **axes** - T2 (optional):
 - Shape: D[dim]
 - Selects the split axis; supports axes 0~3, supports multiple axes simultaneously; when omitted, all axes are processed.
- **steps** - T2 (optional):
 - Shape: D[dim]
 - Per-axis split step sizes. When height*width==1 and channel%subc==0 and starts[1]%subc==0 and ends[1]%subc==0 and steps[1]<=subc, steps can be non-1; otherwise, the NPU can only run when steps==[1,1,1,1].

OUTPUTS

- **output** - T :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T in (tensor<int8>, tensor<float16>): T2 in (tensor<int64>):

Pad

Pads the input tensor with the specified pads and mode.

FORMULA

```
output = Pad(data, pads, mode, constant_value)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::constant_pad_nd
```

INPUTS

- **data** - `T1` (required):
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
- **pads** - `T2` (required):
 - Shape: PADS[batch_begin, channel_begin, height_begin, width_begin, batch_end, channel_end, height_end, width_end]
- **constant_value** - `T1` (optional):
 - Shape: S[scalar]
 - Quantization: per-layer
 - Same type as data; only effective in constant mode, defaults to 0.

OUTPUTS

- **output** - `T1`:
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

ATTRIBUTES

- **mode** - **string**:
 - constant mode has no range restrictions; edge mode only supports h/w directions.

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>` , `tensor<int16>` , `tensor<float16>` , `tensor<bfloat16>`): **T2** in (`tensor<int64>`):

SPECIAL CASES

Reflect Pad In reflect mode, input shape components have additional range restrictions.

- **Input Differences** — **data**: channel range: [1, 8192]; height range: [1, 8192]; width range: [1, 8176]

DepthToSpace

Rearranges the channel dimension of the input tensor back to H/W spatial dimensions by blocksize.

FORMULA

```
N, C, H, W = X.shape
b = blocksize
out_channel = C // (b * b)
Y = np.empty((N, out_channel, H * b, W * b), dtype=X.dtype)

for n in range(N):
    for c in range(out_channel):
        for h in range(H * b):
            for w in range(W * b):
                offset = (h % b) * b + (w % b)
                in_c = c + offset * out_channel
                Y[n, c, h, w] = X[n, in_c, h // b, w // b]
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::pixel_shuffle
```

INPUTS

- **input** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - channel must be divisible by blocksize * blocksize.

OUTPUTS

- **output** - `T1` :
 - Shape: NCHW[batch, out_channel, out_height, out_width]
 - Quantization: per-layer
 - out_channel = channel / (blocksize * blocksize).
 - out_height = height * blocksize.
 - out_width = width * blocksize.

ATTRIBUTES

- **blocksize** - `int`:
 - Spatial block size; must be greater than 0.

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<int16>`, `tensor<float16>`):

SpaceToDepth

Rearranges H/W spatial blocks of the input tensor to the channel dimension.

FORMULA

```
N, C, H, W = X.shape
b = blocksize
Y = np.empty((N, C * b * b, H // b, W // b), dtype=X.dtype)

for n in range(N):
    for c in range(C):
        for h in range(H):
            for w in range(W):
                out_c = c + ((h % b) * b + (w % b)) * C
                Y[n, out_c, h // b, w // b] = X[n, c, h, w]
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::pixel_unshuffle
```

INPUTS

- **input** - **T1**:
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - height and width must be divisible by blocksize.

OUTPUTS

- **output** - **T1**:
 - Shape: NCHW[batch, out_channel, out_height, out_width]
 - Quantization: per-layer
 - out_channel = channel * blocksize * blocksize.
out_height = height / blocksize.
out_width = width / blocksize.

ATTRIBUTES

- **blocksize** - **int**:
 - Spatial block size; must be greater than 0.

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<int16>`, `tensor<float16>`):

Resize

Resizes the spatial dimensions of the input tensor according to the specified scales or sizes, supporting nearest and linear interpolation modes.

FORMULA

```
Y = Resize(X, scales, mode, coordinate_transformation_mode, nearest_mode)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::upsample_bilinear2d, aten::upsample_nearest2d
```

INPUTS

- **X** - `T1` (required):
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
- **roi** - `T3` (optional):
 - Shape: N[8]
 - Crop region coordinates [start_x, start_y, start_z, start_t, end_x, end_y, end_z, end_t]; only the default value [0,0,0,0,1,1,1,1] is supported.
- **scales** - `T3` (optional):
 - Shape: N[4]
 - Per-dimension scale factors [batch_scale, channel_scale, height_scale, width_scale], mutually exclusive with sizes. In linear and nearest modes, only integer scaling is supported, and batch/channel dimension scale must be 1, H/W dimension scale ≤ 8 .
- **sizes** - `T4` (optional):
 - Shape: N[4]
 - Target output size [batch, channel, height, width], mutually exclusive with scales. In linear and nearest modes, only integer scaling is supported, batch/channel dimensions remain unchanged, and H/W dimension multiplier ≤ 8 .

OUTPUTS

- **Y - T1 :**
 - Shape: NCHW[batch, channel, out_height, out_width]
 - Quantization: per-layer

ATTRIBUTES

- **mode - string:**
 - linear and nearest modes are internally implemented via convolution simulation.
- **coordinate_transformation_mode - string:**
 - tf_crop_and_resize is not supported.
- **cubic_coeff_a - float:**
 - Cubic mode coefficient; currently only supports the default value -0.75.
- **exclude_outside - int:**
 - Excluding outer points is not supported.
- **extrapolation_value - float:**
 - Extrapolation value setting is not supported.
- **nearest_mode - string:**
 - Rounding method in nearest mode.
- **antialias - int:**
 - Antialiasing is not supported.
- **keep_aspect_ratio_policy - string:**
 - Only the default stretch mode is supported.
- **axes - int[axis_0, axis_1, axis_2, axis_3]:**
 - Specifies the scaling dimension; the current implementation does not impose additional constraints on this attribute.

DATA TYPE CONSTRAINTS

T1 in (tensor<int8> , tensor<float16>): **T3** in (tensor<float32>): **T4** in (tensor<int64>):

exWindow

exWindow is derived from the window_partition and window_reverse operations in https://github.com/microsoft/Swin-Transformer/blob/main/models/swin_transformer.py . In the original paper, the input feature is divided into multiple non-overlapping windows, each of size window_size × window_size. Both input and output features must be 4D shapes, and division is performed along both height and width dimensions.

FORMULA

```
B, C, H, W = X.shape

if mode == "partition":
    x = X.reshape(B, C, num_windows[0], window_sizes[0],
                  num_windows[1], window_sizes[1])
    y = np.transpose(x, (0, 1, 2, 4, 3, 5))
    Y = y.reshape(B, C,
                  num_windows[0] * num_windows[1],
                  window_sizes[0] * window_sizes[1])

elif mode == "reverse":
    x = X.reshape(B, C, num_windows[0], num_windows[1],
                  window_sizes[0], window_sizes[1])
    y = np.transpose(x, (0, 1, 2, 4, 3, 5))
    Y = y.reshape(B, C,
                  num_windows[0] * window_sizes[0],
                  num_windows[1] * window_sizes[1])

elif mode == "partition_num_first":
    x = X.reshape(B, C, window_sizes[0], num_windows[0],
                  window_sizes[1], num_windows[1])
    y = np.transpose(x, (0, 1, 3, 5, 2, 4))
    Y = y.reshape(B, C,
                  num_windows[0] * num_windows[1],
                  window_sizes[0] * window_sizes[1])
```

OPERATOR LEVEL

RKNN3 Extension Operator

INPUTS

- **X - T1**:
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - partition requires height = num_h * win_h, width = num_w * win_w.
reverse requires height = num_h * num_w, width = win_h * win_w.
partition_num_first requires height = win_h * num_h, width = win_w * num_w.

OUTPUTS

- **Y - T1**:
 - Shape: NCHW[batch, channel, out_height, out_width]
 - Quantization: per-layer
 - partition and partition_num_first output out_height = num_h * num_w, out_width = win_h * win_w.
reverse outputs out_height = num_h * win_h, out_width = num_w * win_w.

ATTRIBUTES

- **mode - string:**
 - partition follows the window_partition implementation.
reverse follows the window_reverse implementation.
The difference between partition_num_first and partition is that partition_num_first divides H/W by num_windows, while partition divides H/W by window_sizes.
- **num_windows - int[num_h, num_w]:**
 - num_h range: [1, 8192]
 - num_w range: [1, 8192]
 - 2D vector, number of windows. Defines how many windows the height and width of the input feature are divided into.
- **window_sizes - int[win_h, win_w]:**
 - win_h range: [1, 8192]
 - win_w range: [1, 8192]
 - 2D vector, window size. Defines the height and width of each window. Attention computation and feature extraction within the window are based on this size.

DATA TYPE CONSTRAINTS

T1 in (tensor<int8> , tensor<float16>):

Index & Scatter

Gather

Gathers elements from the input tensor along the specified axis based on indices.

FORMULA

```
output = Gather(input, indices, axis)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::embedding , aten::gather
```

INPUTS

- **input - T1 :**
 - Shape: NCHW[batch, channel, height, width]

- Quantization: per-layer
- **indices** - `T2` :
 - Shape: INDICES[batch, index]

OUTPUTS

- **output** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

ATTRIBUTES

- **axis** - `int`:
 - Gathers elements along this axis.

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>` , `tensor<float16>`): **T2** in (`tensor<int32>`):

GatherElements

Gathers elements from the input tensor along the specified axis by indices; the output shape matches that of indices.

FORMULA

```
output = GatherElements(data, indices, axis)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::gather
```

INPUTS

- **data** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
- **indices** - `T2` :
 - Shape: NCHW[batch, channel, height, width]
 - Supports both constant and non-constant indices.

OUTPUTS

- **output** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

ATTRIBUTES

- **axis** - `int`:
 - Gathers elements along this axis; negative values are normalized to the valid range.

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<float16>`, `tensor<int32>`): **T2** in (`tensor<int32>`):

GatherND

Gathers slices from the input tensor based on indices and combines them into a new output tensor.

FORMULA

```
output = GatherND(data, indices, batch_dims)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::gather_nd
```

INPUTS

- **data** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
- **indices** - `T2` :
 - Shape: NCHW[batch, channel, height, width]

OUTPUTS

- **output** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

ATTRIBUTES

- **batch_dims** - int:
 - Specifies how many leading dimensions are treated as batch dimensions; no gather operation is performed on these dimensions.

DATA TYPE CONSTRAINTS

T1 in (tensor<int8> , tensor<float16>): **T2** in (tensor<int32>):

ScatterND

Scatters updates into the data tensor at positions specified by indices to produce an output tensor, supporting reduction modes.

FORMULA

```
output = ScatterND(data, indices, updates, reduction)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::scatter_nd
```

INPUTS

- **data** - T1 :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
- **indices** - T2 :
 - Shape: INDICES[num_index, index_dim]
- **updates** - T1 :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

OUTPUTS

- **output** - T1 :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

ATTRIBUTES

- **reduction - string:**
 - Specifies the reduction mode for scatter.

DATA TYPE CONSTRAINTS

T1 in (tensor<int8> , tensor<float16>): **T2** in (tensor<int32> , tensor<int64>):

Normalization

BatchNormalization

Performs batch normalization on the input tensor (ref: <https://arxiv.org/abs/1502.03167>).

FORMULA

```
output = (input - mean) / sqrt(variance + epsilon)
```

OPERATOR LEVEL

Standard ONNX Operator

PYTORCH SOURCE

```
aten::batch_norm
```

INPUTS

- **X** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
- **scale** - **T2** :
 - Shape: C[channel]
 - Quantization: per-channel
- **B** - **T2** :
 - Shape: C[channel]
 - Quantization: per-channel
- **mean** - **T2** :
 - Shape: C[channel]
 - Quantization: per-channel
 - During inference, it is folded into scale/B together with var and does not participate in computation at runtime.
- **var** - **T2** :

- Shape: C[channel]
- Quantization: per-channel
- During inference, it is folded into scale/B together with mean and does not participate in computation at runtime.

OUTPUTS

- **output** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

ATTRIBUTES

- **epsilon** - float:
 - A small real number added to the standard deviation to prevent division by zero; default 1e-5.

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>` , `tensor<float16>`): **T2** in (`const_tensor<float16>`):

BatchNormRelu

Fused BatchNormalization and Relu computation.

FORMULA

```
output = Relu(BatchNormalization(input))
```

INHERITANCE

Inherits from [BatchNormalization](#).

OPERATOR LEVEL

Fused Operator

PYTORCH SOURCE

```
aten::batch_norm
```

exNorm

Performs extended normalization computation, with optional gamma scaling and beta shift.

FORMULA

```
x = X
weight = W if W is not None else 1
bias = B if B is not None else 0

if mode == "layer":
    x_centered = x - np.mean(x, axis=axes, keepdims=True)
    y = x_centered * fast_inv_sqrt(
        np.mean(x_centered * x_centered, axis=axes, keepdims=True) + eps)

elif mode == "mean_var":
    mean = np.mean(x, axis=axes, keepdims=True)
    var = np.mean(x * x, axis=axes, keepdims=True) - mean * mean
    y = (x - mean) * fast_inv_sqrt(var + eps)

elif mode == "rms":
    y = x * fast_inv_sqrt(
        np.mean(x * x, axis=axes, keepdims=True) + eps)

elif mode == "pixel":
    y = x * fast_inv_sqrt(
        np.sum(x * x, axis=axes, keepdims=True) + eps)

elif mode == "l1":
    y = x / np.maximum(
        np.sum(np.abs(x), axis=axes, keepdims=True), eps)

elif mode == "l2":
    y = x / np.maximum(
        fast_sqrt(np.sum(x * x, axis=axes, keepdims=True)), eps)

elif mode == "rms_reciprocal":
    y = fast_inv_sqrt(
        np.mean(x * x, axis=axes, keepdims=True) + eps)

elif mode == "gaussian_topk":
    mean = np.mean(x, axis=axes, keepdims=True)
    std = fast_sqrt(np.mean(x * x, axis=axes, keepdims=True) - mean * mean)
    cutoff_x = mean + std * std_multipler
    y = np.clip(x - cutoff_x, a_min=0, a_max=None).astype(x.dtype)

Y = y * weight + bias
```

OPERATOR LEVEL

RKNN3 Extension Operator

PYTORCH SOURCE

```
aten::layer_norm, aten::rms_norm
```

INPUTS

- **X** - `T1` (required):
 - Shape: NCHW[batch, channel, height, width]

- Quantization: per-layer
- **W** - **T1** (optional):
 - Shape: NCHW[batch, channel, height, width], [1]
 - Quantization: per-layer
 - Gamma scale parameter; supports per-layer, per-channel, or broadcast shapes compatible with the output.
- **B** - **T1** (optional):
 - Shape: NCHW[batch, channel, height, width], [1]
 - Quantization: per-layer
 - Beta shift parameter; supports per-layer, per-channel, or broadcast shapes compatible with the output.

OUTPUTS

- **Y** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - When mode is rms_reciprocal, the dimensions specified by axes maintain keepdims semantics and output as 1.

ATTRIBUTES

- **mode - string:**
 - Valid values: layer, mean_var, rms, pixel, l1, l2, rms_reciprocal, gaussian_topk.
- **axes - int[axis0, axis1, axis2]:**
 - Default value is [1].
Shape inference supports negative axes; the NPU subgraph implementation primarily supports [1] and [1, 2, 3].
- **eps - float:**
 - Default value 1e-5, used to avoid division by zero.
- **std_multiplier - float:**
 - Only used by gaussian_topk; default value is 1.0.

DATA TYPE CONSTRAINTS

T1 in (**tensor<int8>** , **tensor<float16>** , **tensor<bfloat16>**):

exSoftmax

Applies softmax normalization along the specified axis of the input; the output shape is the same as the input.

FORMULA

```
exSoftmax(input, axis) = exp(input - max(input, axis)) / sum_axis(exp(input - max(input, axis)))
```

OPERATOR LEVEL

RKNN3 Extension Operator

PYTORCH SOURCE

```
aten::softmax
```

INPUTS

- **input** - `T1` :
 - Shape: NCHW[batch, channel, height, width]
 - Subject to the constraints of transpose specifications.

OUTPUTS

- **output** - `T1` :
 - Shape: NCHW[batch, channel, height, width]

ATTRIBUTES

- **axis** - `int`:
 - The axis for softmax; 1 is the channel direction, 3 is the width direction. Supporting only axis 1 and 3 is an RKNN platform limitation; the ONNX standard allows any valid axis.

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<float16>`):

exSoftmaxMask

Masked Softmax computation: sets masked positions to -inf using mask_value, then applies Softmax.

FORMULA

```
input(x, mask_t, mask_value) = x + (mask_t - 1) * inf, if mask_value = 0  
x + mask_t * (-inf), if mask_value = 1
```

```
exSoftmaxMask(input, axis) = exp(input - max(input, axis)) / sum_axis(exp(input - max(input, axis)))
```

OPERATOR LEVEL

RKNN3 Extension Operator

PYTORCH SOURCE

```
aten::softmax
```

INPUTS

- **input** - `T1` (required):
 - Shape: NCHW[batch, channel, height, width]
 - Subject to the constraints of transpose specifications.
- **mask** - `T1` (optional):
 - Shape: NCHW[1, channel, height, width]
 - channel range: [1, 8192]
 - width range: [1, 8192]
 - When axis=1, width has no restriction; when axis=3/-1, width range is [1, 8192].

OUTPUTS

- **output** - `T1` :
 - Shape: NCHW[batch, channel, height, width]

ATTRIBUTES

- **axis** - `int`:
 - The axis for softmax; 1 is the channel direction, 3 is the width direction. Supporting only axis 1 and 3 is an RKNN platform limitation; the ONNX standard allows any valid axis.
- **mask_value** - `int`:
 - 0 indicates positions to keep, 1 indicates positions to mask.

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<float16>`):

Matrix Operations

exMatMul

Matrix multiplication operation.

FORMULA

$Y = A \cdot B + C$, where C is bias (bias is optional)

OPERATOR LEVEL

RKNN3 Extension Operator

PYTORCH SOURCE

`aten::matmul`, `aten::bmm`, `aten::mm`

INPUTS

- **A** - `T1` (required):
 - Shape: NK1N[batch, k, 1, n]
 - Quantization: per-layer, per-channel, group32/64/128
- **B** - `T1` (required):
 - Shape: NK1M[batch, k, 1, m]
 - Quantization: per-layer
- **C** - `T1`, `T2` (optional):
 - Shape: NM1M[batch, n, 1, m]
 - Quantization: per-layer
 - Supports NumPy-style broadcasting; broadcast constraints are the same as the Expand operator.

OUTPUTS

- **Y** - `T1`:
 - Shape: NM1M[batch, n, 1, m]
 - Quantization: per-layer

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>`, `tensor<float16>`): **T2** in (`const_tensor<float16>`):

exMatMulActivation

Fused exMatMul and generic Activation computation; activation type is determined by the Activation sub-operator.

FORMULA

```
output = Activation(exMatMul(input, weight, bias))
```

INHERITANCE

Inherits from [exMatMul](#).

OPERATOR LEVEL

RKNN3 Extension Operator

PYTORCH SOURCE

```
aten::matmul, aten::bmm, aten::mm
```

LLM Extension Operators

exAttention

General attention subgraph operator supporting FullAttention and SlidingAttention modes.

FORMULA

```
output = Matmul(Softmax(Matmul(rope(query, cos_pos_emb, sin_pos_emb), rope(key,
cos_pos_emb, sin_pos_emb)) + attention_mask), value)
output = Matmul(Softmax(Matmul(query, key) + attention_mask), value)
```

OPERATOR LEVEL

RKNN3 Extension Operator

PYTORCH SOURCE

```
aten::multi_head_attention_forward
```

INPUTS

- **query** - T1 (required):
 - Shape: NCHW[num_attention_heads, head_dim, 1, seq_len]
- **key** - T1 (required):
 - Shape: NCHW[num_key_value_heads, head_dim, 1, seq_len]
- **value** - T1 (required):
 - Shape: NCHW[num_key_value_heads, head_dim, 1, seq_len]
- **attention_mask** - T1 (required):
 - Shape: NCHW[1, seq_len, 1, seq_len]
- **cos_pos_emb** - T1 (optional):

- Shape: NCHW[1, head_dim, 1, max_position_embeddings]
- **sin_pos_emb** - T1 (optional):
 - Shape: NCHW[1, head_dim, 1, max_position_embeddings]
- **Th** - T2 (required):
 - Shape: N[1]
 - history token len; the number of tokens previously processed.
- **Tc** - T2 (required):
 - Shape: N[1]
 - current token len; the number of valid tokens in the current inference bucket.
- **Ts** - T2 (required):
 - Shape: N[1]
 - system prompt token len; the aligned system prompt length. Both [182X, 3572] require alignment to a multiple of 32.
- **Tsr** - T2 (required):
 - Shape: N[1]
 - Real system prompt token len; the actual system prompt length, less than or equal to Ts.

OUTPUTS

- **output** - T1:
 - Shape: NCHW[num_attention_heads, head_dim, 1, seq_len]

ATTRIBUTES

- **attention_type** - string:
 - Represents global attention and sliding window attention modes respectively. Valid values: [FullAttention, SlidingAttention], default FullAttention.
- **rope** - int:
 - Whether to use RoPE positional encoding; 0 means not used, 1 means used, default 0.
- **kvcache_store_method** - string:
 - Indicates the KV cache storage method; Normal stores float16 data directly, GroupQuant stores with group quantization. Valid values: [Normal, GroupQuant], default Normal.
- **kvcache_dtype** - string:
 - Indicates the KV cache data type. Valid values: [Float16, Int4_to_F8, Int4_to_F16]. Float16 uses float16 storage; Int4_to_F8 uses 4-bit integer quantization storage and converts to 8-bit float during computation; Int4_to_F16 uses 4-bit integer quantization storage and converts to 16-bit float during computation. Default Float16.

- **kvcache_buffer_len - int:**
 - Maximum length of KV cache, in tokens; must be less than the model's max_position_embeddings.
- **kvcache_group_size - int:**
 - Effective when kvcache_store_method is GroupQuant, indicating the group size for group quantization; when kvcache_store_method is Normal, defaults to 32.
- **kvcache_residual_depth - int:**
 - Effective when kvcache_store_method is GroupQuant, indicating that group quantization is triggered after storing this number of tokens; when kvcache_store_method is Normal, defaults to 32.
- **sliding_window_size - int:**
 - Effective when attention_type is SlidingAttention, indicating the sliding window size; when attention_type is FullAttention, defaults to -1.

DATA TYPE CONSTRAINTS

T1 in (tensor<float16>): **T2** in (tensor<int32>):

exSDPAttention

Performs scaled dot-product attention with optional mask.

FORMULA

```
q = Query
k = Key
v = Value

score = MatMul(k, q) * scale
if Mask is not None:
    score = score + Mask

prob = Softmax(score, axis=1)
Y = MatMul(v, prob)
```

OPERATOR LEVEL

RKNN3 Extension Operator

PYTORCH SOURCE

```
aten::scaled_dot_product_attention
```

INPUTS

- **Query** - **T1** (required):

- Shape: NCHW[batch, query_length, head, query_channel]
- Quantization: per-layer
- q matrix input, corresponding to query in the code.
- **Key** - `T1` (required):
 - Shape: NCHW[batch, key_length, head, key_channel]
 - Quantization: per-layer
 - k matrix input, corresponding to key in the code; participates in qk_matmul.
- **Value** - `T1` (required):
 - Shape: NCHW[batch, value_length, head, value_channel]
 - Quantization: per-layer
 - v matrix input.
- **Mask** - `T1` (optional):
 - Shape: NCHW[batch, key_channel, head, query_channel]
 - Quantization: per-layer
 - attention mask; when present, it is added to the qk_matmul result before softmax. The shape must be broadcastable to the qk_matmul output.

OUTPUTS

- **SDPAOutput** - `T1` :
 - Shape: NCHW[batch, query_length, head, value_channel]
 - Quantization: per-layer
 - Final output of scaled dot-product attention.

ATTRIBUTES

- **scale - float:**
 - Default value 1.0, applied to the qk_matmul result.
- **QMatmul - string:**
 - Quantization parameter description for qk_matmul output.
- **QAdd - string:**
 - Quantization parameter description for mask add output; only used when Mask is present.
- **QSoftmax - string:**
 - Quantization parameter description for softmax output.

DATA TYPE CONSTRAINTS

T1 in (`tensor<int8>` , `tensor<float16>` , `tensor<bfloat16>`):

exLinearAttention

Chunk Gated Delta Rule linear attention subgraph operator; computes recursive linear attention of query/key/value in chunks.

FORMULA

```
output = LinearAttention(query, key, value, g, beta, initial_state)
```

OPERATOR LEVEL

RKNN3 Extension Operator

INPUTS

- **query** - `T1` (required):
 - Shape: HD1S[H, D, 1, S]
 - Quantization: per-layer
 - H is num_heads, D is head_dim, S is seq_len.
- **key** - `T1` (required):
 - Shape: HD1S[H, D, 1, S]
 - Quantization: per-layer
- **value** - `T1` (required):
 - Shape: HD1S[H, D, 1, S]
 - Quantization: per-layer
- **g** - `T1` (required):
 - Shape: 1H1S[1, H, 1, S]
 - Quantization: per-layer
 - Gate factor.
- **beta** - `T1` (required):
 - Shape: 1H1S[1, H, 1, S]
 - Quantization: per-layer
- **Tc** - `T2` (required):
 - Shape: S[seq_len]
 - Effective seq_len scalar. When $Tc < seq_len$, left-side zero padding is applied to query/key/value along the seq dimension.
- **initial_state** - `T1` (optional):
 - Shape: HDD[H, D, 1, D]
 - Quantization: per-layer
 - Optional initial recurrent state. If provided by the frontend, it is directly connected.

OUTPUTS

- **output** - `T1`:
 - Shape: `HD1S[H, D, 1, S]`
 - Quantization: per-layer
- **final_state** - `T1`:
 - Shape: `HDD[H, D, 1, D]`
 - Quantization: per-layer
 - Only output when `OutputFinalState` is true.

ATTRIBUTES

- **ChunkSize** - `int`:
 - Chunk size, default 32. `seq_len` must be a multiple of `ChunkSize` (padding is the caller's responsibility).
- **UseQkL2Norm** - `bool`:
 - Whether to apply L2 normalization to query/key; default false.
- **OutputFinalState** - `bool`:
 - Whether to output the final recurrent state; default false.

DATA TYPE CONSTRAINTS

T1 in (`tensor<float16>` , `tensor<bfloat16>`): **T2** in (`tensor<int32>`):

NOTES

Subgraph operator; the actual implementation is expanded by `createSubGraph` at the IR layer; `emit` only returns success.

The batch dimension is fixed to 1. `seq_len` must be padded to a multiple of `ChunkSize` before entering the subgraph.

exRoPE

General subgraph operator supporting RoPE positional encoding.

FORMULA

```
output = input * cos_pos_emb + rotate_half(input) * sin_pos_emb
```

OPERATOR LEVEL

RKNN3 Extension Operator

INPUTS

- **input** - **T1** (required):
 - Shape: NCHW[batch_size, channel, 1, seq_len]
- **cos_pos_emb** - **T1** (required):
 - Shape: NCHW[batch_size, channel, 1, position_embeddings]
 - position_embeddings == 1 or seq_len <= position_embeddings <= max_position_embeddings。
- **sin_pos_emb** - **T1** (required):
 - Shape: NCHW[batch_size, channel, 1, position_embeddings]
 - position_embeddings == 1 or seq_len <= position_embeddings <= max_position_embeddings。
- **Th** - **T2** (optional):
 - Shape: N[1]
 - history token len; the number of tokens previously processed.
- **Tc** - **T2** (optional):
 - Shape: N[1]
 - current token len; the number of valid tokens in the current inference bucket.
- **Ts** - **T2** (optional):
 - Shape: N[1]
 - system prompt token len; the aligned system prompt length. Both [182X, 3572] require alignment to a multiple of 32.
- **Tsr** - **T2** (optional):
 - Shape: N[1]
 - Real system prompt token len; the actual system prompt length, less than or equal to Ts.

OUTPUTS

- **output** - **T1** :
 - Shape: NCHW[batch_size, channel, 1, seq_len]

DATA TYPE CONSTRAINTS

T1 in (`tensor<float16>`): **T2** in (`tensor<int32>`):

NOTES

When exRoPE is used in an LLM model and full cos/sin_pos_emb is provided (position_embeddings > seq_len), inputs such as Th/Tc/Ts/Tsr will be present. exRoPE internally calculates the actual length of cos/sin_pos_emb to use based on

Th/Tc/Ts/Tsr, thereby supporting dynamic sequence length inference. In other cases, seq_len == position_embeddings and Th/Tc/Ts/Tsr do not exist.

exSum

Weighted sum of multiple inputs, commonly used for merging LoRA multi-branch results.

FORMULA

```
Y = Sum(X_i * scale_i) for i in 0..N-1
```

OPERATOR LEVEL

RKNN3 Extension Operator

PYTORCH SOURCE

```
aten::sum
```

INPUTS

- **X** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer
 - Variable-length input list; at least 1, all input shapes must be identical.

OUTPUTS

- **Y** - **T1** :
 - Shape: NCHW[batch, channel, height, width]
 - Quantization: per-layer

ATTRIBUTES

- **scales** - **float[scale_1, ..., scale_N-1]**:
 - Weighting coefficient for each extra input (except the first); the count is the number of inputs - 1.

DATA TYPE CONSTRAINTS

T1 in (**tensor<float16>**):

I/O Processing

exInputProc

Performs input data preprocessing.

OPERATOR LEVEL

RKNN3 Extension Operator

INPUTS

- **A** - **T1** :
 - Shape: NCHW[batch, channel, height, width]

OUTPUTS

- **B** - **T2** :
 - Shape: NCHW[batch, channel, height, width]

DATA TYPE CONSTRAINTS

T1 in (tensor<float32> , tensor<float16> , tensor<bool> , tensor<uint8>): **T2** in (tensor<float16> , tensor<int8>):

exOutputProc

Performs output data post-processing.

OPERATOR LEVEL

RKNN3 Extension Operator

INPUTS

- **A** - **T1** :
 - Shape: NCHW[batch, channel, height, width]

OUTPUTS

- **B** - **T2** :
 - Shape: NCHW[batch, channel, height, width]

DATA TYPE CONSTRAINTS

T1 in (tensor<float16> , tensor<int8>): **T2** in (tensor<float32> , tensor<float16> , tensor<bool> , tensor<uint8>):